MCV4U

Lines and Planes

Website coursebook

20 October 2026

This coursebook follows the classroom sequence in Chapters 8 and 9. Start with line equations in two dimensions, develop plane equations in three dimensions, and then solve intersections, systems and shortest-distance problems.

Work through the written lessons and examples, complete the four homework sets, and use the quizzes before attempting the unit tests and final examination. Every homework problem and assessment question has a typed solution. Written-response questions include model answers and marking criteria.

The online website contains the video lesson slots, interactive three-dimensional visualizer and calculators. The group's original video recordings will be added to those slots. This PDF preserves the written instructional and assessment content.

Open the website

https://mcv4u-lines-and-planes.hozantaxprivate.chatgpt.site

Assessment categories

Knowledge and Understanding (K), Application (A), Thinking (T), and Communication (C).

How to use solutions

Attempt a question first. Compare your method with the worked solution and check your conclusion in the original equations. Multiple-choice answers have a fixed key. Written marks are a guide for self-assessment and may differ from a teacher's awarded mark.

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1 Video lesson sequence

The teaching team prepares four recordings aligned with the written lessons. The durations below are production targets. Open the website to watch each recording after it has been uploaded.

1.1 Video 01 Line equations in two and three dimensions

Sections 8.1, 8.2, 8.3. 12 mins planned.

1. Use one anchor and a nonzero direction to describe a line.
2. Convert between vector, parametric, Cartesian and symmetric forms where applicable.
3. Handle zero direction components and verify points with one shared parameter.

1.2 Video 02 Plane equations and sketches

Sections 8.4, 8.5, 8.6. 12 mins planned.

1. Construct a plane from valid geometric information.
2. Connect independent spans and a perpendicular normal to the equation forms.
3. Sketch intercepts and interpret a missing variable without inventing an intercept.

1.3 Video 03 Intersections and systems

Sections 9.1, 9.2, 9.3, 9.4. 15 mins planned.

1. Classify line and plane relationships by solving and checking all coordinates.
2. Use elimination to distinguish unique, infinite and inconsistent solution sets.
3. Interpret the common solution of two or three plane equations geometrically.

1.4 Video 04 Shortest distance to lines and planes

Sections 9.5, 9.6. 10 mins planned.

1. Choose the distance method for the object and dimension.
2. Keep radical results exact before giving a decimal approximation.
3. Verify that a perpendicular foot lies on the target line or plane.

2 Lessons

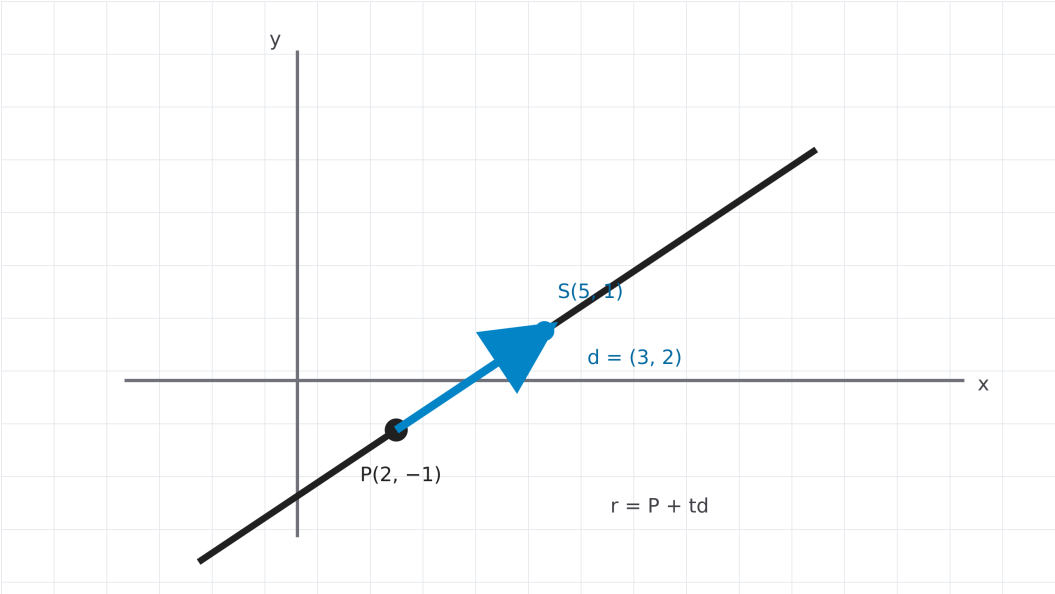
2.1 8.1 Vector and parametric equations of a line in R²

Describe a line using one known point, one non-zero direction vector, and one real parameter. Test whether a point belongs to the line.

Classroom reference: Chapter 8 slides, PDF pages 3, 5, 6, 9, 10, 11, 12, 13.

Assigned textbook practice: Textbook p. 433: 2, 3, 4, 5, 8, 9, 10, 13(a), 14

A position vector locates the starting point. Adding a scalar multiple of a direction vector reaches every point on the line, in either direction.



A line in two dimensions uses one anchor and one direction

Method

1. Choose a point on the line as the anchor.
2. Subtract two given points to obtain a non-zero direction vector, or use the direction supplied.
3. Write the vector equation, then separate its x- and y-components.
4. To test a point, check that one value of the parameter satisfies both coordinates.

Equations and notation

Vector equation

$$\vec{r} = (x_0, y_0) + t(a, b), \qquad t \in \mathbb{R}, \quad (a, b) \neq (0, 0)$$

The anchor is (x_0,y_0), and (a,b) is a direction vector.

Parametric equations

$$x = x_0 + at, \qquad y = y_0 + bt$$

The same parameter controls both coordinates.

Direction through two points

$$\vec{d} = \overrightarrow{PQ} = (x_Q - x_P,\ y_Q - y_P)$$

Any non-zero scalar multiple gives the same line.

Perpendicular directions

$$(a, b) \cdot (-b, a) = 0$$

Rotate a non-zero direction through 90 degrees to obtain a perpendicular direction.

Common mistake

A line extends in both directions. The parameter is any real number unless a line segment or restricted motion is specified. Do not use a different parameter for each coordinate.

Check your understanding

1. Substituting t=0 returns the anchor.
2. A second known point must be produced by one shared parameter.
3. Different anchors or scaled directions can describe the same line.

Worked example

Write vector and parametric equations for the line through P(2,-1) and Q(8,3). Then determine whether S(5,1) lies on it.

1 Find the direction. Subtract P from Q and simplify by a non-zero factor.

$$\overrightarrow{PQ} = (6, 4) = 2(3, 2)$$

2 Write both forms. Use P as the anchor and (3,2) as the direction.

$$\vec{r} = (2, -1) + t(3, 2), \quad x = 2 + 3t, \quad y = -1 + 2t$$

3 Test S. The x-coordinate requires t=1. Check that value in y.

$$5 = 2 + 3t \Rightarrow t = 1, \qquad -1 + 2(1) = 1$$

4 Verify the original data. The simplified direction reaches Q when t=2.

$$(2, -1) + 2(3, 2) = (8, 3)$$

Conclusion

S lies on the line because one parameter, t=1, satisfies both coordinates.

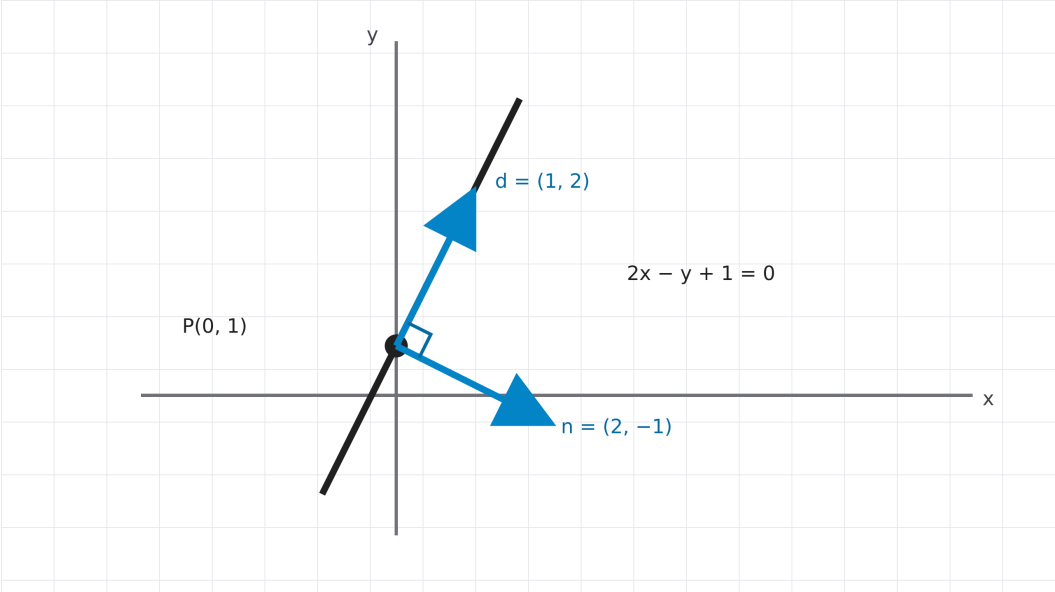
2.2 8.2 Cartesian equations and conversions in R²

Connect Cartesian, vector, and parametric forms using normal and direction vectors. Recognise vertical lines and calculate the acute angle between lines.

Classroom reference: Chapter 8 slides, PDF pages 14, 15, 17, 18, 19, 20, 21, 22, 23, 24, 25.

Assigned textbook practice: Textbook p. 443: 1(a,b,e), 3, 4, 5, 6, 7, 8, 9(b), 10(a,b), 11(b)

In a Cartesian equation, the coefficients give a normal vector. A line direction is perpendicular to that normal. Converting forms changes the notation, not the set of points.



A Cartesian line has perpendicular direction and normal vectors

Method

1. Read the normal from the coefficients of x and y.
2. Choose a perpendicular direction and find a convenient point that satisfies the equation.
3. Build vector and parametric forms, or eliminate the parameter to recover Cartesian form.
4. For an acute angle, use the absolute value of the dot product of two directions or two normals.

Equations and notation

Cartesian equation and directions

$$Ax + By + C = 0, \quad (A, B) \neq (0, 0), \quad \vec{n} = (A, B), \quad \vec{d} = (B, -A)$$

A normal is perpendicular to every displacement along the line.

From vector to Cartesian form

$$\vec{n} \cdot \big((x, y) - (x_0, y_0)\big) = 0$$

Choose a non-zero normal perpendicular to the supplied direction.

Symmetric form in R²

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} \quad (ab \neq 0)$$

Solve both parametric equations for the same parameter. If a direction component is zero, keep that coordinate as a constant equation instead.

Vertical and horizontal lines

$$x = c \quad \text{or} \quad y = c$$

A zero direction component means that coordinate remains constant.

Acute angle between lines

$$\theta = \cos^{-1}\left(\frac{|\vec{d_1} \cdot \vec{d_2}|}{\|\vec{d_1}\|\|\vec{d_2}\|}\right), \quad 0^\circ \leq \theta \leq 90^\circ$$

Both directions must be non-zero. Normals may be used in place of directions.

Common mistake

The vector (A,B) is a normal to Ax+By+C=0, not a direction. Vertical lines cannot be written with a finite slope, but vector and Cartesian forms still work.

Check your understanding

1. The anchor satisfies the Cartesian equation.
2. The direction has zero dot product with the normal.
3. Multiplying every term of a Cartesian equation by a non-zero constant preserves the line.

Worked example

Convert 2x-3y+6=0 to vector and parametric form, then verify the conversion.

1 Read a normal and choose a direction. The direction must be perpendicular to the normal.

$$\vec{n} = (2, -3), \qquad \vec{d} = (3, 2), \qquad \vec{n} \cdot \vec{d} = 6 - 6 = 0$$

2 Find an anchor. Set x=0 and solve for y.

$$-3y + 6 = 0 \Rightarrow P = (0, 2)$$

3 Write the line. Insert the anchor and direction.

$$\vec{r} = (0, 2) + t(3, 2), \qquad x = 3t, \quad y = 2 + 2t$$

4 Substitute to verify. Every parameter value must satisfy the original equation.

$$2(3t) - 3(2 + 2t) + 6 = 0$$

Conclusion

The Cartesian, vector, and parametric equations describe the same line.

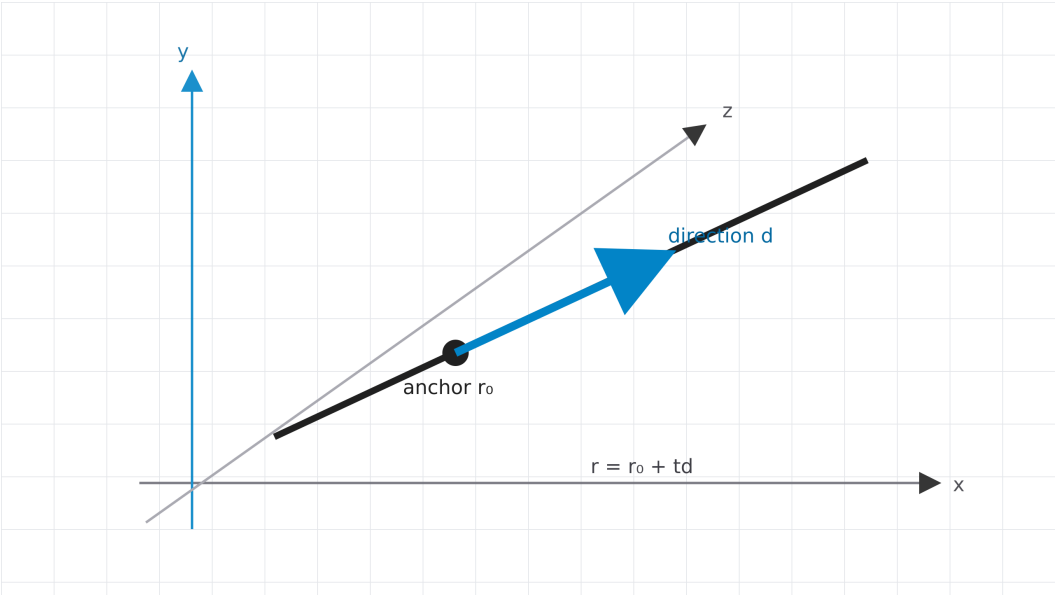
2.3 8.3 Vector parametric and symmetric equations in R³

Extend line equations to three coordinates, convert to symmetric form, and recognise equivalent representations and zero direction components.

Classroom reference: Chapter 8 slides, PDF pages 26, 27, 28, 29, 30, 31, 32, 33, 36, 37.

Assigned textbook practice: Textbook p. 449: 1(a,b,c), 3, 5(a,e,f), 6, 8(a), 9, 10(a,b), 15

A line in three-space still has only one independent direction and one parameter. Three coordinates do not mean three independent parameters.



A line starts at an anchor point and follows a direction vector

Method

1. Find an anchor and a non-zero direction in three dimensions.
2. Separate the vector equation into three parametric equations.
3. Solve the non-constant coordinate equations for the shared parameter to obtain symmetric form.
4. When a direction component is zero, write the corresponding coordinate as a separate constant equation.

Equations and notation

Vector and parametric forms

$$\vec{r} = (x_0, y_0, z_0) + t(a, b, c), \quad x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$$

Use one real parameter and a direction that is not the zero vector.

Symmetric form

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \qquad (abc \neq 0)$$

These expressions all equal the same parameter.

A zero direction component

$$b = 0 \quad \Rightarrow \quad y = y_0, \qquad \frac{x - x_0}{a} = \frac{z - z_0}{c} \quad (ac \neq 0)$$

If two components are zero, keep two constant-coordinate equations and the remaining coordinate free.

Equivalent line test

$$\vec{d_2} = k\vec{d_1}\ (k \neq 0), \qquad \vec{r}_{02} \in L_1$$

Parallel directions and one shared point establish that the lines coincide.

Common mistake

Never divide by a zero direction component in symmetric form. A single non-degenerate linear Cartesian equation in R³ describes a plane; a Cartesian line needs two independent plane constraints.

Check your understanding

1. The same parameter must reproduce x, y, and z.
2. Parallel directions alone do not prove that two equations describe the same line.
3. A negative scalar multiple reverses the parameter direction without changing the line.

Worked example

Find all three forms of the line through A(1,2,-1) and B(5,2,5). Explain how to handle the unchanged y-coordinate.

1 Find a direction. Subtract A from B and simplify.

$$\overrightarrow{AB} = (4, 0, 6) = 2(2, 0, 3)$$

2 Vector and parametric forms. Anchor at A and retain one shared parameter.

$$\vec{r} = (1, 2, -1) + t(2, 0, 3), \quad x = 1 + 2t, \quad y = 2, \quad z = -1 + 3t$$

3 Symmetric form. Only the x- and z-equations can be divided by their direction components.

$$\frac{x-1}{2} = \frac{z+1}{3}, \qquad y = 2$$

4 Check B. Use t=2 in all coordinates.

$$(1, 2, -1) + 2(2, 0, 3) = (5, 2, 5)$$

Conclusion

The line lies in the plane y=2, but it is only a line because x and z depend on one shared parameter.

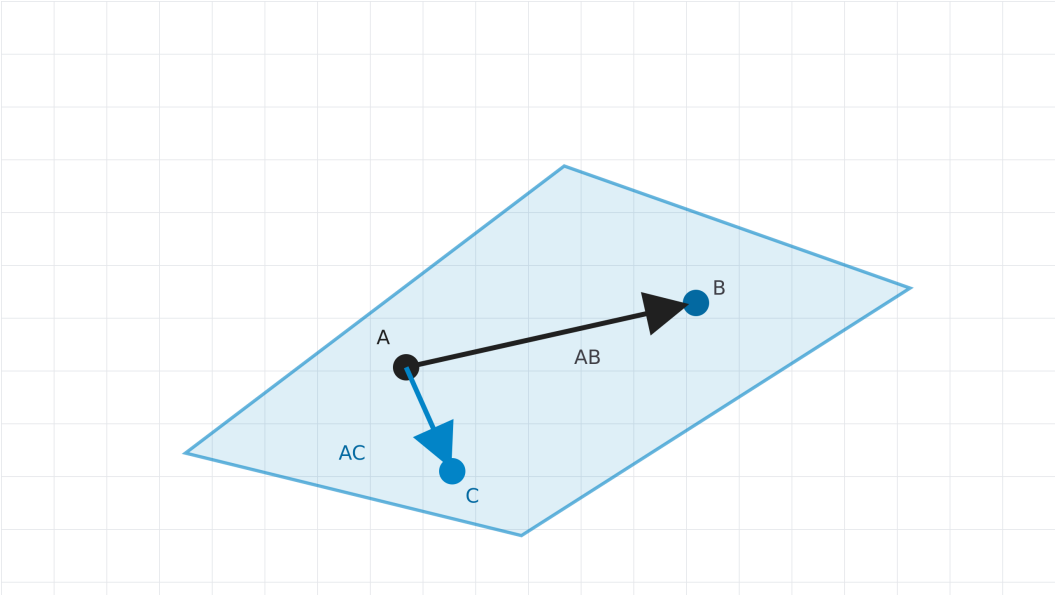
2.4 8.4 Vector and parametric equations of a plane

Construct planes from points and lines. Explain why two independent spanning directions are necessary and identify the four standard geometric definitions.

Classroom reference: Chapter 8 slides, PDF pages 34, 35, 38, 39, 40, 41, 42, 43, 44, 45.

Assigned textbook practice: Textbook p. 459: 1, 2, 4, 5, 6(b), 7, 9, 10, 11, 12, 13(a)

A plane requires an anchor and two independent directions. These directions can come from three non-collinear points, two intersecting lines, two distinct parallel lines, or a line and a point outside it.



A unique plane needs two independent directions

Method

1. Choose an anchor on the plane.
2. Build two in-plane vectors from the supplied points or lines.
3. Verify that the vectors are not parallel.
4. Write a vector equation with two independent parameters, then separate its coordinates.

Equations and notation

Vector and parametric plane

$$\vec{r} = \vec{p} + s\vec{a} + t\vec{b}, \quad \vec{a} \times \vec{b} \neq \vec{0}, \qquad x = p_1 + sa_1 + tb_1,\ y = p_2 + sa_2 + tb_2,\ z = p_3 + sa_3 + tb_3$$

Both parameters range independently over the real numbers.

Three non-collinear points

$$\vec{p} = \overrightarrow{OA}, \qquad \vec{a} = \overrightarrow{AB}, \qquad \vec{b} = \overrightarrow{AC}$$

The connecting vectors must not be scalar multiples.

Intersecting or parallel lines

$$\vec{r} = \vec{p} + s\vec{d_1} + t\vec{d_2} \quad \text{or} \quad \vec{r} = \vec{p} + s\vec{d} + t(\vec{q} - \vec{p})$$

For intersecting distinct lines, p is their intersection. For distinct parallel lines, p and q lie on different lines.

A line and an external point

$$\vec{r} = \vec{p} + s\vec{d} + t(\vec{P} - \vec{p})$$

The external point P must not lie on the given line.

Common mistake

Two skew lines are not coplanar. Two coincident lines provide only one independent direction, and three collinear points do not determine a unique plane.

Check your understanding

1. Two independent parameters are needed for a plane.
2. Each given point must be reproducible by some parameter pair.
3. Check the cross product of the spans is non-zero before claiming a unique plane.

Worked example

Find vector and parametric equations for the plane through A(1,0,2), B(2,1,2), and C(1,1,3).

1 Build two spans. Use A as the common starting point.

$$\vec{a} = \overrightarrow{AB} = (1,1,0), \qquad \vec{b} = \overrightarrow{AC} = (0,1,1)$$

2 Check independence. The vectors are not scalar multiples; their cross product is non-zero.

$$\vec{a} \times \vec{b} = (1,-1,1) \neq \vec{0}$$

3 Write vector form. Add independent multiples of both spans to A.

$$\vec{r} = (1,0,2) + s(1,1,0) + t(0,1,1), \qquad s,t \in \mathbb{R}$$

4 Separate coordinates. Read the contribution of each parameter to each coordinate.

$$x = 1 + s, \qquad y = s + t, \qquad z = 2 + t$$

Conclusion

The parameter pairs (0,0), (1,0), and (0,1) reproduce A, B, and C respectively.

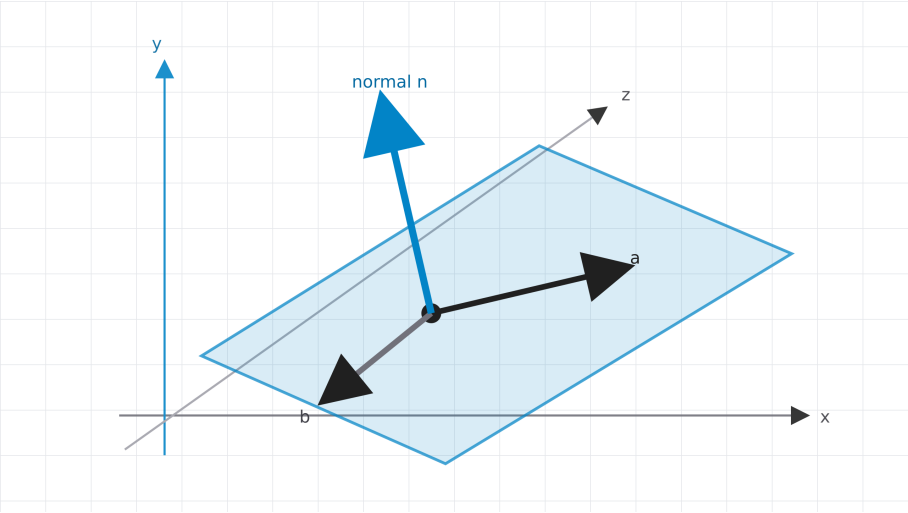
2.5 8.5 Cartesian equations and properties of planes

Use normal vectors to construct Cartesian equations, convert between plane forms, and compare parallel, perpendicular, and intersecting planes.

Classroom reference: Chapter 8 slides, PDF pages 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58.

Assigned textbook practice: Textbook p. 468: 1, 3, 5, 6(a), 7, 8, 9(a), 10, 11, 12, 13, 15

A plane normal is perpendicular to every displacement inside the plane. The Cartesian coefficients record this normal directly.



Two in-plane directions span the plane and the normal points out of it

Method

1. Obtain a non-zero normal directly or by crossing two independent spans.
2. Insert a known point into point-normal form and expand.
3. To recover a vector equation, choose one point and two independent vectors perpendicular to the normal.
4. Use the normals to classify plane orientation and calculate the acute angle between planes.

Equations and notation

Point-normal and Cartesian forms

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0 \quad \Longleftrightarrow \quad Ax + By + Cz + D = 0, \qquad \vec{n} = (A, B, C) \neq \vec{0}$$

A normal may be scaled by any non-zero factor if the complete equation is scaled too.

Normal from spans

$$\vec{n} = \vec{a} \times \vec{b}$$

A non-zero cross product is perpendicular to both independent in-plane directions.

Parallel and perpendicular planes

$$\vec{n}_1 \times \vec{n}_2 = \vec{0} \quad \text{or} \quad \vec{n}_1 \cdot \vec{n}_2 = 0$$

The first condition gives parallel orientation. The second gives perpendicular planes.

Acute angle between planes

$$\theta = \cos^{-1}\left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\|\vec{n}_2\|}\right)$$

Use the absolute value so the reported plane angle lies between 0 and 90 degrees.

Common mistake

Parallel normals show that planes are parallel or coincident, but the constants decide which. A zero normal does not define a plane.

Check your understanding

1. The normal has zero dot product with both spans.
2. The anchor satisfies the final Cartesian equation.
3. When converting back to vector form, the new spans must be independent and perpendicular to the normal.

Worked example

Convert r=(1,0,2)+s(1,1,0)+t(0,1,1) to Cartesian form, then recover a vector form from that equation.

1 Find the normal. Cross the two spans.

$$\vec{n} = (1,1,0) \times (0,1,1) = (1,-1,1)$$

2 Use the anchor. Write point-normal form and expand.

$$(x-1) - y + (z-2) = 0 \quad \Rightarrow \quad x - y + z = 3$$

3 Choose free coordinates. Starting from the Cartesian equation, let y=u and z=v.

$$x = 3 + u - v, \qquad y = u, \qquad z = v$$

4 Recover vector form. Separate constants and parameter contributions.

$$\vec{r} = (3,0,0) + u(1,1,0) + v(-1,0,1)$$

Conclusion

Both vector equations describe the same plane x-y+z=3, even though their anchors and spans differ.

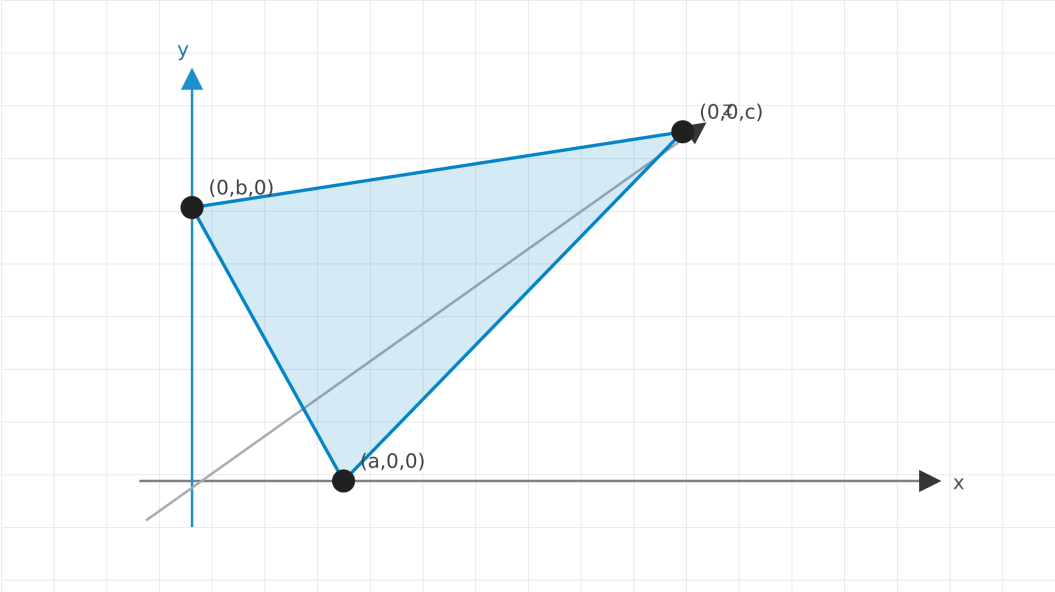
2.6 8.6 Sketching points and planes in R³

Plot ordered triples and use axis intercepts, coordinate-plane traces, or spanning directions to show a plane's orientation.

Classroom reference: Chapter 8 slides, PDF pages 2.

Plane sketching is listed in the Chapter 8 learning goals and required in Topic H. The supplied Chapter 8 slides end with Section 8.5.

A sketch represents an infinite plane with a small patch. Labels and traces carry the mathematical information; the drawing does not need to be a perfect perspective picture.



Three axis intercepts establish the orientation of a plane

Method

1. Draw labelled x-, y-, and z-axes and plot points in the correct coordinate order.
2. Set two coordinates to zero to find each available axis intercept.
3. If intercepts are missing or all meet at the origin, use traces in coordinate planes or two independent directions.
4. Draw a labelled patch and indicate that the plane continues beyond it.

Equations and notation

Axis intercepts when defined

$$Ax + By + Cz = D: \quad (D/A, 0, 0),\ (0, D/B, 0),\ (0, 0, D/C)$$

Each fraction is used only when its denominator is non-zero.

Coordinate-plane traces

$$z = 0:\ Ax + By = D; \qquad y = 0:\ Ax + Cz = D; \qquad x = 0:\ By + Cz = D$$

A trace is the intersection of the plane with a coordinate plane.

A plane parallel to an axis

$$x + y = 2$$

The z-coordinate is free. Sketch the line x+y=2 in the xy-plane and extend it in the z-direction.

A plane through the origin

$$x + y + z = 0 : \quad \vec{r} = s(1, -1, 0) + t(1, 0, -1)$$

The intercept triangle collapses at the origin, so use traces or two independent spans.

Common mistake

A missing coefficient does not always mean that an entire axis is excluded. If the corresponding intercept equation becomes 0=0, the whole axis lies in the plane; if it becomes a contradiction, the plane misses that axis.

Check your understanding

1. Substitute each labelled intercept into the original equation.
2. A plane patch is not a bounded triangle; the plane extends indefinitely.
3. A coefficient of zero means translation in that coordinate direction leaves the equation unchanged.

Worked example

Sketch 2x+3y+6z=6 and explain what would change when sketching x+y=2.

1 Find the x-intercept. Set y=z=0.

$$2x = 6 \Rightarrow A = (3, 0, 0)$$

2 Find the y- and z-intercepts. Set the other two coordinates to zero each time.

$$B = (0, 2, 0), \qquad C = (0, 0, 1)$$

3 Draw and verify the patch. Join the three non-collinear intercepts and label the axes. The full plane extends beyond the triangle.

$$2(3) = 6, \qquad 3(2) = 6, \qquad 6(1) = 6$$

4 Handle a missing intercept. For x+y=2, the z-axis would require 0=2, so there is no z-intercept. The plane is parallel to the z-axis.

$$\vec{r} = (2, 0, 0) + s(-1, 1, 0) + t(0, 0, 1)$$

Conclusion

Use an intercept triangle when three non-collinear intercepts exist. Otherwise use traces or spanning directions.

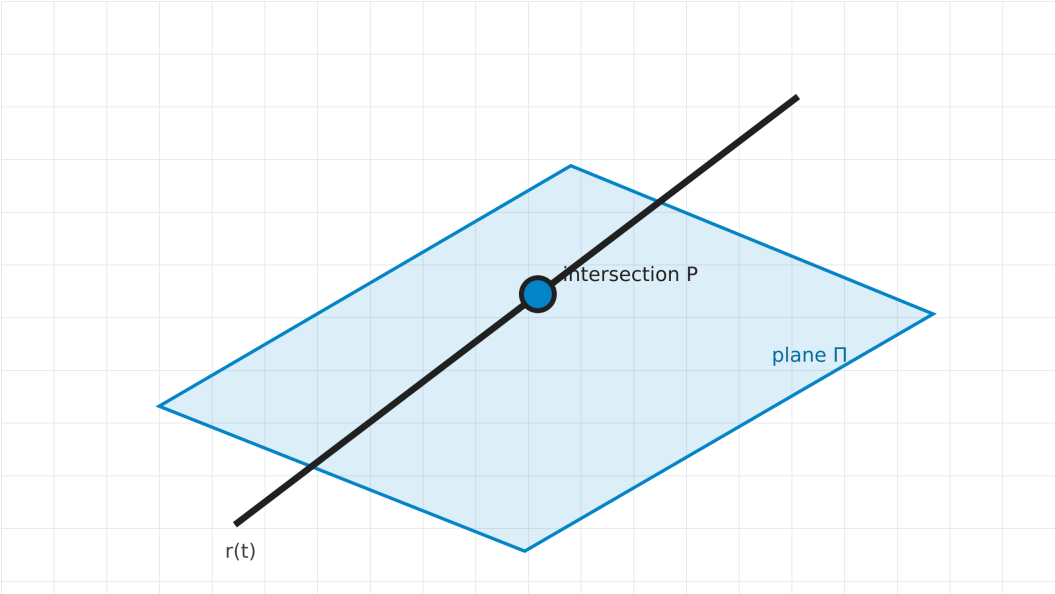
2.7 9.1 Intersections of lines and planes

Classify a line against a plane and compare two lines in R^3, including intersecting, parallel, coincident, and skew cases.

Classroom reference: Chapter 9 slides, PDF pages 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

Assigned textbook practice: Textbook p. 496: 4(a), 5(b), 7, 8(a), 9(a,b), 10, 14(a), 15(a)

An intersection point must satisfy every equation involved. A line-plane problem uses substitution; a two-line problem uses independent parameters and coordinate equations.



The shared point is found by substituting the line into the plane

Method

1. For a line and plane, substitute all parametric coordinates into the plane and solve for the parameter.
2. Interpret one parameter value, an identity, or a contradiction as one point, a contained line, or no intersection.
3. For two lines, use different parameters and equate corresponding coordinates.
4. Check every coordinate and compare directions before classifying the lines.

Equations and notation

A line and a plane

$$\vec{r} = \vec{p} + t\vec{d}, \quad \vec{n} \cdot \vec{r} = k \quad \Rightarrow \quad (\vec{n} \cdot \vec{d})t = k - \vec{n} \cdot \vec{p}$$

A non-zero parameter coefficient gives one intersection. With a zero coefficient, compare the constants.

Contained or disjoint

$$0t = 0 \Rightarrow L \subset \Pi; \qquad 0t = c\ (c \neq 0) \Rightarrow L \cap \Pi = \varnothing$$

The second case is a line parallel to and outside the plane.

Two-line intersection system

$$\vec{p} + s\vec{d_1} = \vec{q} + t\vec{d_2}$$

Solve the coordinate equations for s and t and verify all three.

Two-line classification

$$\vec{d_1} \parallel \vec{d_2}: \text{ coincident or distinct parallel}; \qquad \vec{d_1} \nparallel \vec{d_2}: \text{ intersecting or skew}$$

For parallel directions, test whether an anchor lies on the other line. For non-parallel directions, solve the coordinate system.

Common mistake

Two non-parallel lines need not intersect in R³. If their coordinate system has no solution and their directions are not parallel, the lines are skew. Solving only two coordinate equations is not enough.

Check your understanding

1. The final point must satisfy both objects.
2. An identity means all parameter values work, not just t=0.
3. Skew lines are neither parallel nor intersecting and cannot lie in one plane.

Worked example

Find where r=(1,0,2)+t(1,2,-1) meets x+y+z=5. Then find where r=(0,0,0)+s(1,1,0) meets r=(1,0,0)+u(-1,1,0).

1 Substitute the first line. Write its coordinates in the plane equation.

$$(1 + t) + 2t + (2 - t) = 5 \Rightarrow t = 1$$

2 Find and check the point. Return to the line, then check the plane.

$$P = (2, 2, 1), \qquad 2 + 2 + 1 = 5$$

3 Set up the two-line system. Use separate parameters for the two lines.

$$s = 1 - u, \qquad s = u, \qquad 0 = 0$$

4 Solve and verify. The first two equations give s=u=1/2, and the third coordinate agrees.

$$Q = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Conclusion

The first line crosses its plane at P. The two lines intersect at Q because the same parameter pair satisfies all three coordinate equations.

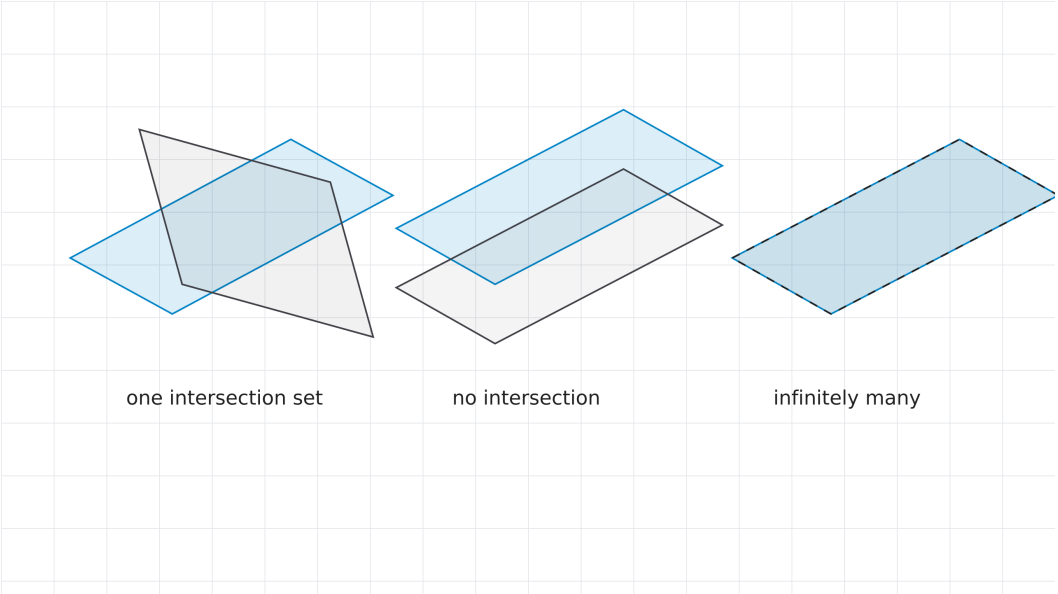
2.8 9.2 Solving systems of linear equations

Use elimination and substitution for up to three equations in three unknowns. Distinguish a unique solution, infinitely many solutions, and an inconsistent system.

Classroom reference: Chapter 9 slides, PDF pages 13, 14, 15, 16, 17.

Assigned textbook practice: Textbook p. 508: 5, 12(a,b,c)

Elimination replaces equations with equivalent restrictions. A contradiction means no common point; an identity removes a repeated restriction; free variables describe all remaining solutions.



Algebraic outcomes match geometric relationships

Method

1. Label the equations and eliminate one variable from two equation pairs.
2. Solve the resulting smaller system, keeping signs and constants aligned.
3. Back-substitute and verify every original equation.
4. If elimination produces an identity or contradiction, state the solution family or explain why no solution exists.

Equations and notation

Unique solution

$$x = a, \qquad y = b, \qquad z = c$$

A consistent system has a unique solution when all three variables are determined.

An identity

$$0 = 0$$

This equation adds no restriction. Use the remaining equations to determine whether free variables remain.

A contradiction

$$0 = c \qquad (c \neq 0)$$

The system is inconsistent and has no solution.

An infinite solution family

$$x + y + z = 3,\ y = 1 \quad \Rightarrow \quad (x, y, z) = (2 - t, 1, t)$$

One free variable gives a line of solutions; two free variables can give a plane.

Common mistake

The equation 0=0 does not prove that every triple is a solution. Keep the other non-redundant equations and express any free variables with parameters.

Check your understanding

1. Multiply every term, including the right side, during elimination.
2. Consistent means at least one solution, including an infinite solution set.
3. A proposed solution must satisfy every original equation.

Worked example

Solve x+y+z=6, 2x-y+z=3, and x+2y-z=2 by elimination.

1 Eliminate x. Subtract twice equation 1 from equation 2, and equation 1 from equation 3.

$$-3y - z = -9, \qquad y - 2z = -4$$

2 Eliminate y. Add three times the second reduced equation to the first.

$$(-3y - z) + 3(y - 2z) = -9 + 3(-4) \Rightarrow -7z = -21 \Rightarrow z = 3$$

3 Back-substitute. Use the reduced equation to find y, then the first original equation to find x.

$$y - 6 = -4 \Rightarrow y = 2, \qquad x + 2 + 3 = 6 \Rightarrow x = 1$$

4 Verify. Check all three original equations.

$$1 + 2 + 3 = 6, \qquad 2 - 2 + 3 = 3, \qquad 1 + 4 - 3 = 2$$

Conclusion

The system is consistent with the unique solution (1,2,3).

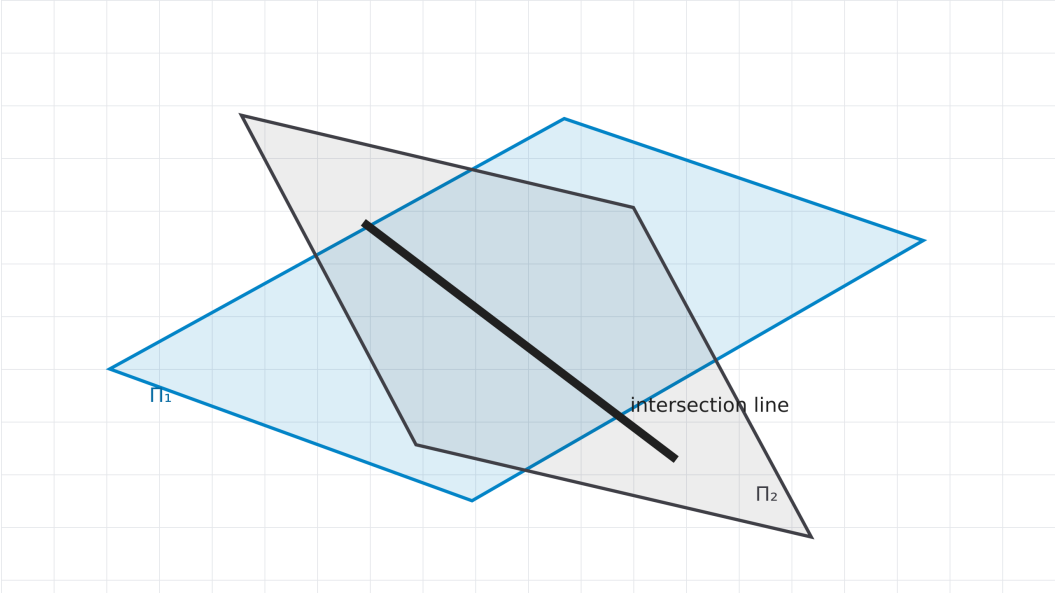
2.9 9.3 Intersections of two planes

Compare plane normals and solve the common system to identify an intersection line, coincident planes, or distinct parallel planes.

Classroom reference: Chapter 9 slides, PDF pages 18, 19, 20, 21, 22, 23, 24, 25, 26.

Assigned textbook practice: Textbook p. 516: 5(a,b), 6 (find the solution when one exists), 9, 11(a)

Two non-parallel planes meet along a line. The line direction is perpendicular to both normals, but one point on both planes is also needed.



Non-parallel planes meet along a line

Method

1. Compare the normals and the complete equations.
2. For non-parallel planes, eliminate one variable and choose a free parameter.
3. Write a vector equation for the common points.
4. Verify the anchor satisfies both planes and the direction is perpendicular to both normals.

Equations and notation

Intersection direction

$$\vec{d} = \vec{n}_1 \times \vec{n}_2 \neq \vec{0}$$

A non-zero cross product establishes non-parallel planes and a line of intersection.

Parallel normals

$$\vec{n}_2 = k\vec{n}_1 \quad (k \neq 0)$$

Scale the entire first equation, including its constant, before comparing with the second.

Coincident or distinct

$$\Pi_1 : Ax+By+Cz = D, \quad \Pi_2 : kAx+kBy+kCz = E, \qquad E = kD \Rightarrow \Pi_1 = \Pi_2, \quad E \neq kD \Rightarrow \Pi_1 \cap \Pi_2 = \varnothing$$

Here k is non-zero. Matching constants give the same plane. Different constants give distinct parallel planes with no intersection.

Common mistake

A cross product gives only the direction of the intersection line. It does not supply the anchor point. Proportional normals alone do not distinguish coincident from distinct parallel planes.

Check your understanding

1. The line anchor lies on both planes.
2. Its direction has zero dot product with both normals.
3. One free parameter describes a line, not a single intersection point.

Worked example

Find a vector equation for the intersection of x+y+z=3 and x-y+z=1.

1 Find y. Subtract the second equation from the first to eliminate x and z.

$$2y = 2 \Rightarrow y = 1$$

2 Choose a free variable. Substitute y=1, then let z=t.

$$x + z = 2 \Rightarrow x = 2 - t, \quad y = 1, \quad z = t$$

3 Write the intersection line. Separate the constant point and direction.

$$\vec{r} = (2, 1, 0) + t(-1, 0, 1), \qquad t \in \mathbb{R}$$

4 Check its direction. Cross the two normals and compare.

$$(1, 1, 1) \times (1, -1, 1) = (2, 0, -2) = -2(-1, 0, 1)$$

Conclusion

The planes intersect in the line r=(2,1,0)+t(-1,0,1).

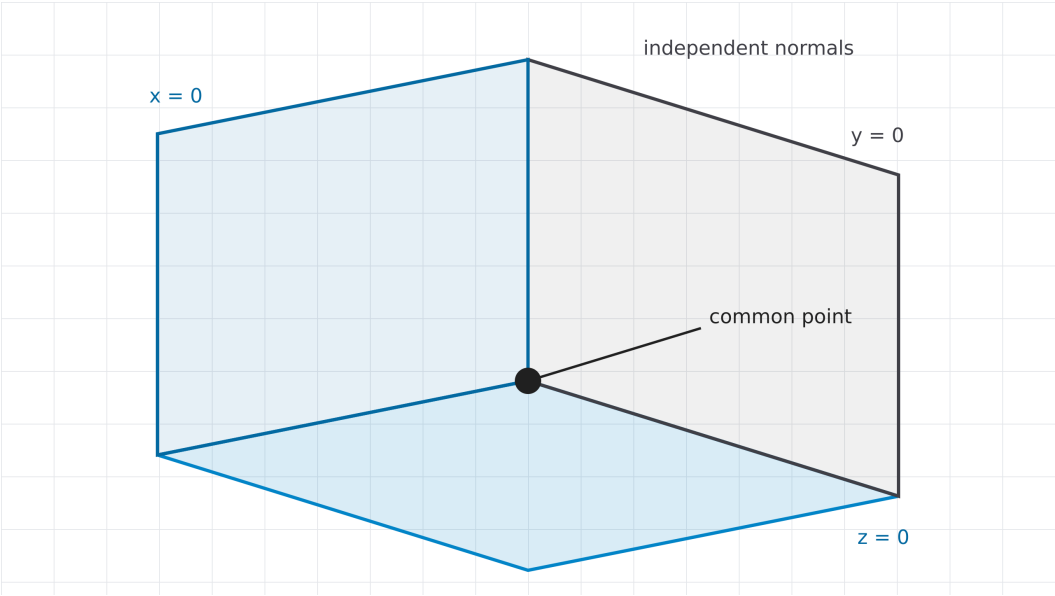
2.10 9.4 Intersections of three planes

Solve a three-plane system and distinguish a common point, common line, common plane, or no common intersection.

Classroom reference: Chapter 9 slides, PDF pages 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37.

Assigned textbook practice: Textbook p. 531: 8, 9, 12, 13

The answer is the set of points shared by all three planes. Pairwise intersections do not guarantee one common point or line across the complete system.



Three independent plane constraints can determine one common point

Method

1. Use elimination to reduce the three Cartesian equations.
2. Look for a contradiction first.
3. If the system is consistent, count the remaining free variables and express the full solution set.
4. Describe the common geometry and verify it in all three original planes.

Equations and notation

A common point

$$x = a, \quad y = b, \quad z = c$$

Three independent restrictions determine one ordered triple.

A common line

$$\vec{r} = \vec{p} + t\vec{d}$$

One free variable remains after consistent elimination.

A common plane

$$\vec{r} = \vec{p} + s\vec{a} + t\vec{b}$$

Two free variables remain when all three equations describe the same plane.

No common intersection

$$0 = c \quad (c \neq 0)$$

A contradiction rules out every common solution, even if some plane pairs intersect.

Common mistake

Three planes can intersect pairwise in three parallel lines while sharing no common point. Do not conclude consistency from pairwise intersections or from non-parallel normals alone.

Check your understanding

1. Use all three equations in the final verification.
2. A redundant equation is not automatically a contradiction.
3. Always state whether the common set is a point, a line, a plane, or empty.

Worked example

Determine the common intersection of x+y+z=6, x-y+z=2, and x+y-z=0.

1 Compare equations 1 and 2. Subtract equation 2 from equation 1.

$$2y = 4 \Rightarrow y = 2$$

2 Compare equations 1 and 3. Subtract equation 3 from equation 1.

$$2z = 6 \Rightarrow z = 3$$

3 Find x. Substitute into the first equation.

$$x + 2 + 3 = 6 \Rightarrow x = 1$$

4 Verify the common point. All three original equations hold.

$$1 + 2 + 3 = 6, \quad 1 - 2 + 3 = 2, \quad 1 + 2 - 3 = 0$$

Conclusion

The three planes have the unique common point (1,2,3), so the system is consistent.

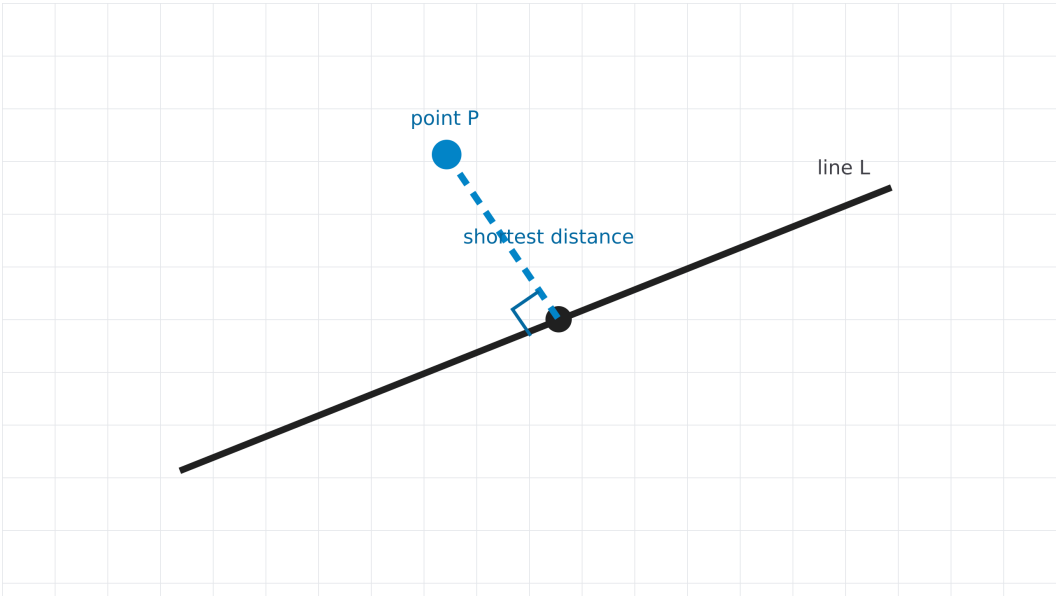
2.11 9.5 Distance from a point to a line in R² and R³

Measure the shortest perpendicular distance using a Cartesian line equation in R² or displacement and direction vectors in R³.

Classroom reference: Chapter 9 slides, PDF pages 38, 39, 40, 41, 42, 43, 44, 45, 46.

Assigned textbook practice: Textbook p. 540: 1(a,b), 2(b), 3(b,c), 5(a), 6(a,b), 7(a)

The displacement from an anchor to the point contains parallel and perpendicular parts. Only the perpendicular part determines the shortest distance.



Distance is measured along the shortest perpendicular segment

Method

1. Identify the dimension and choose the appropriate line representation.
2. In R², substitute the point into the Cartesian line formula.
3. In R³, form the anchor-to-point displacement and use a cross product or projection.
4. Simplify the exact length and state the units; approximate only if requested.

Equations and notation

Point to line in R²

$$d(P, L) = \frac{|Ax_P + By_P + C|}{\sqrt{A^2 + B^2}}, \qquad L : Ax + By + C = 0$$

The denominator is the normal magnitude. A and B cannot both be zero.

Point to line in R³

$$d(P, L) = \frac{\|\overrightarrow{AP} \times \vec{d}\|}{\|\vec{d}\|}$$

A lies on the line, and the non-zero vector d is its direction. The numerator is the parallelogram area.

Projection and foot of the perpendicular

$$t_H = \frac{\overrightarrow{AP} \cdot \vec{d}}{\vec{d} \cdot \vec{d}}, \quad H = A + t_H \vec{d}, \quad d(P, L) = \|P - H\|$$

This method works in both two and three dimensions.

Extension distance between parallel R² lines

$$L_1 : Ax + By + C_1 = 0, \quad L_2 : Ax + By + C_2 = 0 \quad \Rightarrow \quad d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

First make the x- and y-coefficients identical. Alternatively choose a point on one line and find its distance to the other.

Common mistake

The distance from the external point to an arbitrary line anchor is usually longer than the required distance. The final answer is a non-negative scalar, not a vector.

Check your understanding

1. The perpendicular displacement has zero dot product with the line direction.
2. Distance is zero exactly when the point lies on the line.
3. Scaling the direction vector leaves the distance unchanged.

Worked example

Find the distance from P(1,2) to 3x+4y-5=0. Then find the distance from Q(1,0,2) to r=t(1,1,0).

1 Use the R² formula. Substitute P and the Cartesian coefficients.

$$d(P, L) = \frac{|3(1) + 4(2) - 5|}{\sqrt{3^2 + 4^2}} = \frac{6}{5}$$

2 Build the R³ displacement. Use the origin as the second line's anchor.

$$\overrightarrow{AQ} = (1, 0, 2), \qquad \vec{d} = (1, 1, 0)$$

3 Use the cross product. Its magnitude divided by the direction magnitude gives the perpendicular height.

$$\overrightarrow{AQ} \times \vec{d} = (-2, 2, 1), \qquad d(Q, L) = \frac{\sqrt{9}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

4 Check with the perpendicular foot. Project onto the direction and calculate the remaining displacement.

$$t_H = \frac{1}{2}, \quad H = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \quad \|Q - H\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 4} = \frac{3\sqrt{2}}{2}$$

Conclusion

The exact distances are 6/5 units in R^2 and $3\sqrt{2}/2$ units in R^3.

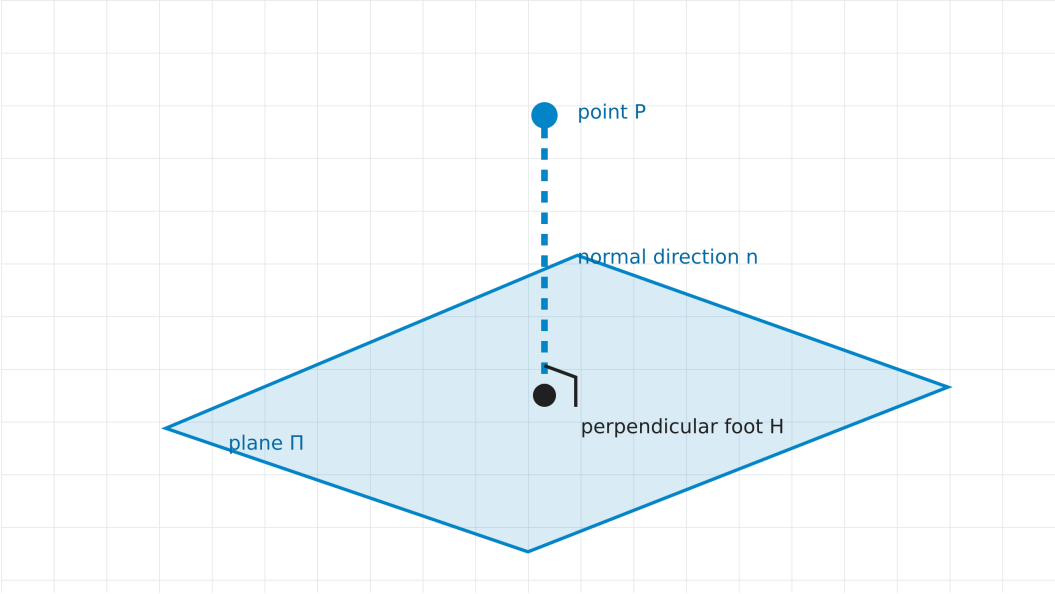
2.12 9.6 Distance from a point to a plane

Use the plane normal to measure a perpendicular distance, find the foot of the perpendicular, and interpret zero or positive distances geometrically.

Classroom reference: Chapter 9 slides, PDF pages 47, 48, 49, 50, 51, 52, 53, 54.

Assigned textbook practice: Textbook p. 549: 2(a,d,e), 3(a), 4(c), 5, 7, 8(a)

The shortest path from a point to a plane is parallel to the plane normal. The scalar equation measures a signed normal displacement before the absolute value turns it into a length.



The shortest path from a point to a plane follows its normal

Method

1. Write the plane as Ax+By+Cz+D=0 and read its non-zero normal.
2. Substitute the point into the numerator and take the absolute value.
3. Divide by the normal magnitude and simplify exactly.
4. If the nearest point is required, move along the normal until the plane equation is satisfied.

Equations and notation

Point to plane

$$d(P,\Pi) = \frac{|Ax_P + By_P + Cz_P + D|}{\sqrt{A^2 + B^2 + C^2}}$$

The normal (A,B,C) must be non-zero.

Projection form

$$d(P,\Pi) = \frac{|\overrightarrow{AP} \cdot \vec{n}|}{\|\vec{n}\|}$$

A is any known point on the plane. This is the magnitude of the scalar projection onto its normal.

Foot of the perpendicular

$$H = P - \frac{Ax_P + By_P + Cz_P + D}{A^2 + B^2 + C^2}(A, B, C)$$

Retain the sign in this expression so the movement is in the correct direction.

Parallel plane extension

$$Ax + By + Cz = D_1, \quad Ax + By + Cz = D_2 \quad \Rightarrow \quad d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$$

First rewrite both planes with exactly the same normal coefficients.

Extension distance between skew lines

$$L_1 : \vec{r} = \vec{A} + s\vec{d_1}, \quad L_2 : \vec{r} = \vec{B} + t\vec{d_2}, \qquad d = \frac{|(\vec{B} - \vec{A}) \cdot (\vec{d_1} \times \vec{d_2})|}{\|\vec{d_1} \times \vec{d_2}\|}$$

Use this when the directions are not parallel. The cross product is a common perpendicular direction. Intersecting non-parallel lines give distance zero.

Common mistake

Use the complete Cartesian expression, including D, and take the absolute value of the entire numerator. The numerator alone is not the distance unless the normal is a unit vector.

Check your understanding

1. A point on the plane has distance zero.
2. The nearest point satisfies the plane equation.
3. The vector from the nearest point to the external point is parallel to the normal.

Worked example

Find the distance from P(3,1,2) to 2x-y+2z-3=0 and find the nearest point H on the plane.

1 Read the normal and substitute. Keep the entire signed numerator together.

$$\vec{n} = (2, -1, 2), \quad 2(3) - 1 + 2(2) - 3 = 6, \quad \|\vec{n}\| = 3$$

2 Calculate the distance. Take the absolute value and divide by the normal magnitude.

$$d = \frac{|6|}{3} = 2$$

3 Find the perpendicular foot. Move from P along the normal by the signed projection factor.

$$H = (3,1,2) - \frac{6}{9}(2,-1,2) = \left(\frac{5}{3}, \frac{5}{3}, \frac{2}{3}\right)$$

4 Verify the geometry. Check that H lies on the plane and PH has length 2.

$$2\left(\frac{5}{3}\right) - \frac{5}{3} + 2\left(\frac{2}{3}\right) - 3 = 0, \qquad \|P - H\| = 2$$

Conclusion

The point is exactly 2 units from the plane, and H=(5/3,5/3,2/3) is the foot of the perpendicular.

3 Homework and worked solutions

3.1 Homework 01 Line representations

Vector, parametric, symmetric, and scalar forms

Suggested time: 40 minutes. Attempt each problem before reviewing its solution.

Problem 1 Through two points

Categories: K, A

Write the line through P(2,-1,3) and Q(-1,5,1) in vector, parametric, and symmetric form.

Hint

Start with Q-P to create a direction vector.

Worked solution

1 Direction. Subtract the points.

$$\vec{d} = Q - P = (-3, 6, -2)$$

2 Vector form. Anchor the line at P.

$$\vec{r} = (2, -1, 3) + t(-3, 6, -2)$$

3 Parametric form. Read the three components.

$$x = 2 - 3t, \quad y = -1 + 6t, \quad z = 3 - 2t$$

4 Symmetric form. Solve each coordinate for t.

$$\frac{x-2}{-3} = \frac{y+1}{6} = \frac{z-3}{-2}$$

Final answer

$\vec{r} = (2, -1, 3) + t(-3, 6, -2)$, with the corresponding forms shown above.

Problem 2 Scalar to vector in R²

Categories: K, T

Convert 2x-3y+6=0 into a vector equation and symmetric equation of the line.

Hint

A direction vector is perpendicular to the scalar line's normal (2,-3).

Worked solution

1 Direction. Rotate the normal by 90°.

$$\vec{n} = (2, -3) \Rightarrow \vec{d} = (3, 2), \quad \vec{n} \cdot \vec{d} = 0$$

2 Anchor. Set x=0 and solve for y.

$$-3y + 6 = 0 \Rightarrow y = 2 \Rightarrow P = (0, 2)$$

3 Write both forms. Use P and d.

$$\vec{r} = (0,2) + t(3,2), \qquad \frac{x}{3} = \frac{y-2}{2}$$

Final answer

$\vec{r} = (0,2) + t(3,2)$ and $x/3 = (y-2)/2$.

Problem 3 Membership test

Categories: A, C

Determine whether S(5,-3,6) lies on r=(1,1,2)+t(2,-1,1). Justify using a shared parameter.

Hint

Solve for t using one coordinate, then check the other two.

Worked solution

1 Use x. Solve the x-coordinate equation.

$$5 = 1 + 2t \Rightarrow t = 2$$

2 Check y. Evaluate the line's y-coordinate at t=2.

$$y = 1 - 2 = -1 \neq -3$$

3 Conclude. The same parameter does not reproduce every coordinate.

$$S \notin L$$

Final answer

No. The parameter forced by x gives y=-1, not y=-3.

Problem 4 Cartesian line in R³

Categories: A, T

Find a vector equation for the line formed by x+y+z=4 and x-y+z=2.

Hint

Eliminate one variable, choose a free parameter, then separate the constant and parameter parts.

Worked solution

1 Eliminate. Subtract the second plane from the first.

$$2y = 2 \Rightarrow y = 1$$

2 Choose a parameter. Both equations now give x+z=3. Let z=t.

$$x = 3 - t, \quad y = 1, \quad z = t$$

3 Write vector form. Separate the fixed point and direction.

$$\vec{r} = (3, 1, 0) + t(-1, 0, 1)$$

Final answer

$\vec{r} = (3, 1, 0) + t(-1, 0, 1)$.

Problem 5 Angle between Cartesian lines in R²

Categories: K, A, C

Find the acute angle between $L_1 : 2x - y = 1$ and $L_2 : x + 3y = 6$. Give an exact inverse-trigonometric expression and an answer to the nearest tenth of a degree.

Hint

Choose a direction perpendicular to each coefficient normal. Use the absolute dot product for the acute angle.

Worked solution

1 Directions. Choose directions perpendicular to the Cartesian normals.

$$\vec{d_1} = (1, 2), \qquad \vec{d_2} = (3, -1)$$

2 Dot product and magnitudes. Use the absolute dot product because the acute angle is requested.

$$|\vec{d_1} \cdot \vec{d_2}| = 1, \qquad \|\vec{d_1}\| = \sqrt{5}, \quad \|\vec{d_2}\| = \sqrt{10}$$

3 Angle. Apply inverse cosine and state the requested precision.

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{50}}\right) \approx 81.9^\circ$$

Final answer

$\theta = \cos^{-1}(1/\sqrt{50}) \approx 81.9^\circ$.

Problem 6 Symmetric form with a fixed coordinate

Categories: A, T, C

Write parametric and symmetric equations for $\vec{r} = (3, -2, 1) + t(0, 2, -3)$. Determine whether $P(3, 2, -5)$ lies on the line.

Hint

The zero x-component of the direction means x stays fixed. Do not divide by zero.

Worked solution

1 Parametric form. Use the same parameter in every coordinate.

$$x = 3, \qquad y = -2 + 2t, \qquad z = 1 - 3t$$

2 Symmetric form. Write the fixed coordinate separately.

$$x = 3, \qquad \frac{y+2}{2} = \frac{z-1}{-3}$$

3 Point membership. The y-coordinate gives t=2. Check both other coordinates.

$$t = 2 \Rightarrow (3, -2+4, 1-6) = (3, 2, -5)$$

Final answer

$x = 3$, $y = -2 + 2t$, $z = 1 - 3t$; equivalently $x = 3$, $(y+2)/2 = (z-1)/(-3)$. The point P lies on the line.

3.2 Homework 02 Building planes

Point-normal, three-point, and line-point definitions

Suggested time: 45 minutes. Attempt each problem before reviewing its solution.

Problem 1 Point and normal

Categories: K, A

Find a Cartesian equation of the plane through P(1,2,-1) with normal n=(2,-1,3).

Hint

Use n · (r-P)=0.

Worked solution

1 Point-normal form. Insert the given point and normal.

$$2(x-1)-(y-2)+3(z+1)=0$$

2 Expand. Distribute and combine constants.

$$2x-y+3z+3=0$$

3 Verify. Substitute P.

$$2(1)-2+3(-1)+3=0$$

Final answer

$2x-y+3z+3=0$.

Problem 2 Three non-collinear points

Categories: A, T

Find the plane through A(1,0,0), B(0,1,0), and C(0,0,2).

Hint

Create AB and AC, then take their cross product.

Worked solution

1 Span vectors. Form two directions from A.

$$\overrightarrow{AB}=(-1,1,0),\quad \overrightarrow{AC}=(-1,0,2)$$

2 Normal. Cross the span vectors.

$$\overrightarrow{AB}\times\overrightarrow{AC}=(2,2,1)$$

3 Scalar equation. Use A in point-normal form.

$$2(x-1)+2y+z=0\Rightarrow 2x+2y+z=2$$

Final answer

$2x+2y+z=2$.

Problem 3 A line and an external point

Categories: T, C

Find the plane containing r=(1,0,2)+t(1,1,0) and the point P(0,1,3).

Hint

Use the line direction and the vector from its anchor to P as two spans.

Worked solution

1 Second span. Connect the line anchor A to P.

$$\overrightarrow{AP} = (-1, 1, 1)$$

2 Normal. Cross the line direction with AP.

$$(1, 1, 0) \times (-1, 1, 1) = (1, -1, 2)$$

3 Plane equation. Use A=(1,0,2).

$$(x - 1) - y + 2(z - 2) = 0 \Rightarrow x - y + 2z = 5$$

Final answer

$x - y + 2z = 5$.

Problem 4 Two intersecting lines

Categories: A, T, C

Two lines intersect at P(2,1,0) and have directions d_1=(1,0,1) and d_2=(0,2,-1). Find their plane.

Hint

The two line directions are spanning vectors. Cross them to find a normal.

Worked solution

1 Find a normal. Cross the two directions.

$$(1, 0, 1) \times (0, 2, -1) = (-2, 1, 2)$$

2 Use the intersection point. Build point-normal form at P.

$$-2(x - 2) + (y - 1) + 2z = 0$$

3 Simplify. Collect the constant terms.

$$-2x + y + 2z = -3$$

Final answer

$-2x + y + 2z = -3$.

Problem 5 Sketching a plane with a missing intercept

Categories: K, A, C

Describe a sketch of $2x-3z=6$. Give the x- and z-intercepts, explain the missing y-intercept, and state how to extend the sketch to represent a plane.

Hint

Set the other two coordinates to zero for each intercept. A missing variable supplies a free direction.

Worked solution

1 Find intercepts. Set z=0 for the x-intercept and x=0 for the z-intercept.

$$X=(3,0,0), \qquad Z=(0,0,-2)$$

2 Test the y-axis. Every y-axis point has x=z=0, which cannot satisfy the equation.

$$x=z=0 \Rightarrow 0=6$$

3 Extend in the free direction. Extend the intercept segment in both y-directions. The points (3,1,0) and (0,1,-2) form a second parallel segment.

$$\vec{r}=(3,0,0)+s(-3,0,-2)+t(0,1,0)$$

Final answer

The intercepts are $(3,0,0)$ and $(0,0,-2)$. There is no y-intercept; the plane extends parallel to the y-axis.

Problem 6 Cartesian to vector and parametric plane equations

Categories: A, T, C

Convert $x-2y+z=4$ into vector and parametric form. Verify the anchor and both spanning vectors.

Hint

Set y=s and z=t, solve for x, then separate the fixed and parameter parts.

Worked solution

1 Choose free parameters. Two independent parameters describe the plane.

$$y=s, \qquad z=t, \qquad x=4+2s-t$$

2 Vector form. Separate the anchor and parameter directions.

$$\vec{r}=(4,0,0)+s(2,1,0)+t(-1,0,1)$$

3 Verify. The anchor satisfies the plane. Both non-parallel spans are perpendicular to the normal.

$$4-0+0=4, \quad (1,-2,1)\cdot(2,1,0)=0, \quad (1,-2,1)\cdot(-1,0,1)=0$$

Final answer

$\vec{r} = (4, 0, 0) + s(2, 1, 0) + t(-1, 0, 1)$, with $x = 4 + 2s - t$, $y = s$, $z = t$, where $s, t \in \mathbb{R}$.

3.3 Homework 03 Intersection systems

Line-plane and plane-plane relationships

Suggested time: 45 minutes. Attempt each problem before reviewing its solution.

Problem 1 Line through a plane

Categories: A

Find the intersection of r=(1,0,2)+t(1,2,-1) and x+y+z=5.

Hint

Substitute the line's three parametric coordinates into the plane.

Worked solution

1 Coordinates. Write the parametric line.

$$x = 1 + t, \quad y = 2t, \quad z = 2 - t$$

2 Solve parameter. Substitute into x+y+z=5.

$$3 + 2t = 5 \Rightarrow t = 1$$

3 Intersection. Evaluate the line at t=1.

$$P = (2, 2, 1)$$

Final answer

One intersection at (2,2,1).

Problem 2 Two planes

Categories: A, T

Find a vector equation for the intersection of x+y+z=3 and x-y+z=1.

Hint

Subtract the equations, then let one remaining variable be a parameter.

Worked solution

1 Eliminate. Subtract the second plane from the first.

$$2y = 2 \Rightarrow y = 1$$

2 Free variable. Both equations now give x+z=2. Let z=t.

$$x = 2 - t, \quad y = 1, \quad z = t$$

3 Vector form. Separate constants and parameter coefficients.

$$\vec{r} = (2, 1, 0) + t(-1, 0, 1)$$

Final answer

$\vec{r} = (2, 1, 0) + t(-1, 0, 1)$.

Problem 3 Infinite intersections

Categories: T, C

Classify the relationship between r=(0,1,2)+t(1,-1,2) and the plane x+y=1.

Hint

Check n · d and whether the line anchor lies in the plane.

Worked solution

1 Parallel test. The plane normal is n=(1,1,0).

$$\vec{n} \cdot \vec{d} = (1,1,0) \cdot (1,-1,2) = 0$$

2 Anchor test. Substitute the anchor (0,1,2).

$$0 + 1 = 1$$

3 Classify. A parallel line with one point in the plane is entirely contained in it.

$$L \subset \Pi$$

Final answer

The line lies in the plane, so there are infinitely many intersection points.

Problem 4 Parallel or contained

Categories: K, T, C

Classify r=(0,0,1)+t(2,-2,0) relative to the plane x+y+z=3. Show both required tests.

Hint

First test n dot d. Then substitute the anchor into the plane.

Worked solution

1 Direction test. The plane normal is n=(1,1,1).

$$\vec{n} \cdot \vec{d} = (1,1,1) \cdot (2,-2,0) = 0$$

2 Anchor test. Evaluate the plane at (0,0,1).

$$0 + 0 + 1 \neq 3$$

3 Classify. The line direction is parallel to the plane, but the anchor is outside.

$$L \parallel \Pi, \qquad L \cap \Pi = \varnothing$$

Final answer

The line is parallel to and disjoint from the plane, so there is no intersection.

Problem 5 Solving a three-variable system by elimination

Categories: A, T, C

Solve $x+y+z=4$, $2x-y+z=8$, and $x+2y-z=-3$ by elimination. Verify your solution and interpret the intersection of the three planes.

Hint

Use the first equation to eliminate x from each other equation.

Worked solution

1 Eliminate x. Subtract twice equation 1 from equation 2, and subtract equation 1 from equation 3.

$$E_2-2E_1:\ -3y-z=0, \qquad E_3-E_1:\ y-2z=-7$$

2 Solve the reduced equations. The first reduced equation gives z=-3y.

$$y-2(-3y)=-7\Rightarrow y=-1, \qquad z=3, \qquad x=2$$

3 Verify. All original equations hold, so the three planes have one common point.

$$2-1+3=4, \qquad 4+1+3=8, \qquad 2-2-3=-3$$

Final answer

The three planes meet at the unique point $(2,-1,3)$.

Problem 6 Three planes with a common line or no common point

Categories: K, T, C

Compare the system $x+y+z=2$, $x-y+z=0$, $2x+2z=2$ with the system formed when the third equation becomes $2x+2z=3$. Give the complete solution set in the consistent case.

Hint

Add the first two equations and compare the result with the third equation in each system.

Worked solution

1 First two equations. Subtract to find y, then substitute into either equation.

$$2y=2\Rightarrow y=1, \qquad x+z=1$$

2 Consistent third plane. The third equation equals the sum of the first two and adds no new restriction.

$$\vec{r}=(1,1,0)+t(-1,0,1), \qquad t\in\mathbb{R}$$

3 Inconsistent third plane. The altered third equation contradicts the sum of the first two.

$$2x+2z=2 \quad \text{and} \quad 2x+2z=3\Rightarrow 0=1$$

Final answer

With third constant 2, the common set is the line $\vec{r} = (1, 1, 0) + t(-1, 0, 1)$. With third constant 3, the system is inconsistent and has no common points.

3.4 Homework 04 Distance and consistency

Shortest distance and geometric system classification

Suggested time: 45 minutes. Attempt each problem before reviewing its solution.

Problem 1 Point-to-line distance in R²

Categories: K, A

Find the distance from P(4,1) to 3x-4y+2=0.

Hint

Use the scalar point-to-line distance formula.

Worked solution

1 Substitute. Insert P into the numerator.

$$|3(4) - 4(1) + 2| = |10| = 10$$

2 Normalize. Find the normal magnitude.

$$\sqrt{3^2 + (-4)^2} = 5$$

3 Distance. Divide.

$$d = \frac{10}{5} = 2$$

Final answer

2 units.

Problem 2 Point-to-line distance in R³

Categories: A, T

Find the distance from P(3,2,0) to r=(1,1,1)+t(2,0,-1).

Hint

Use ||AP × d|| / ||d||.

Worked solution

1 Displacement. Take A=(1,1,1) from the line.

$$\overrightarrow{AP} = (2, 1, -1)$$

2 Cross product. Cross AP with d.

$$\overrightarrow{AP} \times \vec{d} = (-1, 0, -2)$$

3 Distance. Divide magnitudes.

$$d = \frac{\sqrt{5}}{\sqrt{5}} = 1$$

Final answer

1 unit.

Problem 3 Coincident versus parallel

Categories: T, C

Compare the systems formed by P_1: x+2y-z=3 with (a) P_2: 2x+4y-2z=6 and (b) P_3: 2x+4y-2z=7.

Hint

Compare the scale factor for both coefficients and constants.

Worked solution

1 System (a). Every term of P_2 is twice P_1.

$$P_2 = 2P_1 \Rightarrow \text{same plane}$$

2 System (b). The left side scales by 2 but the constant does not.

$$2(3) = 6 \neq 7 \Rightarrow \text{distinct parallel planes}$$

3 Classify. Coincident planes share infinitely many points; distinct parallel planes share none.

$$(a)\ \text{consistent, infinite;} \qquad (b)\ \text{inconsistent}$$

Final answer

(a) coincident and consistent with infinitely many solutions; (b) parallel and inconsistent.

Problem 4 Closest point on a line

Categories: A, T, C

For P(4,2,1) and r=(1,0,1)+t(1,1,0), find the closest point on the line and the exact distance.

Hint

Project AP onto the line direction to find the parameter of the perpendicular foot.

Worked solution

1 Build the displacement. Use the line anchor A=(1,0,1).

$$\overrightarrow{AP} = (3, 2, 0)$$

2 Find the foot parameter. Project AP onto d=(1,1,0).

$$t = \frac{\overrightarrow{AP} \cdot \vec{d}}{\vec{d} \cdot \vec{d}} = \frac{5}{2}$$

3 Find the foot. Evaluate the line at t=5/2.

$$H = \left(\frac{7}{2}, \frac{5}{2}, 1\right)$$

4 Measure the distance. Take the magnitude of PH.

$$d = \left\| \left(\frac{1}{2}, -\frac{1}{2}, 0 \right) \right\| = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2}$$

Final answer

The closest point is $H = (7/2, 5/2, 1)$, and the distance is $\sqrt{2}/2$.

Problem 5 Point to a vector line in R²

Categories: A, T, C

Find the exact distance from $P(5, 1)$ to $L : \vec{r} = (1, -2) + t(3, 4)$. Explain why the length of the connector from the anchor to P is not the required distance.

Hint

Use the normal (4,-3) to obtain a Cartesian equation, then calculate perpendicular distance.

Worked solution

1 Convert the line. The normal (4,-3) is perpendicular to (3,4). Use the anchor (1,-2).

$$4(x - 1) - 3(y + 2) = 0 \Rightarrow 4x - 3y - 10 = 0$$

2 Calculate perpendicular distance. Divide the absolute residual by the normal magnitude.

$$d = \frac{|4(5) - 3(1) - 10|}{\sqrt{4^2 + (-3)^2}} = \frac{7}{5}$$

3 Interpret. The anchor connector is not perpendicular and is longer than the shortest separation.

$$\overrightarrow{AP} = (4, 3), \quad \overrightarrow{AP} \cdot (3, 4) = 24 \neq 0, \quad \|\overrightarrow{AP}\| = 5 > \frac{7}{5}$$

Final answer

The shortest distance is $7/5$ units. The anchor-to-P connector is not perpendicular and is not the shortest connector.

Problem 6 Point to a plane and the perpendicular foot

Categories: K, A, T, C

Find the exact distance from $P(3, 1, 4)$ to $x + 2y + 2z = 4$. Find the perpendicular foot H and verify that the connector is normal to the plane.

Hint

The normal is (1,2,2). Remove the normal displacement from P to locate the foot.

Worked solution

1 Residual and distance. Use the absolute scalar residual and normal magnitude.

$$d = \frac{|3 + 2(1) + 2(4) - 4|}{\sqrt{1 + 4 + 4}} = \frac{9}{3} = 3$$

2 Perpendicular foot. Subtract the required normal displacement from P.

$$H = P - \frac{9}{9}(1,2,2) = (2,-1,2)$$

3 Verify. H lies on the plane and HP is a normal vector.

$$2 + 2(-1) + 2(2) = 4, \qquad P - H = (1,2,2) = \vec{n}$$

Final answer

The distance is 3 units and the perpendicular foot is $H = (2,-1,2)$.

4 Quizzes

4.1 Line equation fluency

Line equations, point membership, line relationships, and mathematical justification.

12 questions 23 marks Suggested time: 30 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: K

For $\vec{r} = (2, -1, 4) + t(3, -2, 5)$, which vector controls the line's orientation?

A. $(2, -1, 4)$

B. $(3, -2, 5)$

C. $(5, -3, 9)$

D. $(3, 2, -5)$

Question 2 [1 mark]

Categories: A

Which scalar equation describes the R² line through (2,-1) with direction d=(4,3)?

A. $4x + 3y - 5 = 0$

B. $3x - 4y - 10 = 0$

C. $3x + 4y - 2 = 0$

D. $4x - 3y - 11 = 0$

Question 3 [1 mark]

Categories: C

How should $\vec{r} = (1, 3, 4) + t(2, 0, -1)$ be written in symmetric form?

A. $\frac{x-1}{2} = \frac{y-3}{0} = \frac{z-4}{-1}$

B. $\frac{x-1}{2} = \frac{z-4}{-1},\ y = 3$

C. $\frac{x+1}{2} = \frac{z+4}{-1},\ y = 0$

D. $x = 1,\ \frac{y-3}{2} = \frac{z-4}{-1}$

Question 4 [1 mark]

Categories: T
The lines $\vec{r} = (1, 0, 2) + t(2, -1, 3)$ and $\vec{r} = (5, -2, 8) + s(4, -2, 6)$ have what relationship?

A. Parallel and distinct

B. The same line

C. Perpendicular

D. Skew

Question 5 [1 mark]

Categories: K, C
Why does one Cartesian equation in R³ not define a line?

A. One linear equation in R³ defines a plane.

B. A line cannot be described with equations.

C. Every Cartesian equation defines a point.

D. A line needs three independent parameters.

Question 6 [1 mark]

Categories: K, A
Find the exact magnitude of the direction vector $\vec{d} = (2, -3, 6)$.

A. 5

B. 7

C. $\sqrt{11}$

D. 11

Question 7 [1 mark]

Categories: A
Which direction vector can be used for the line through $A(-2, 1, 4)$ and $B(4, 3, 0)$?

A. $(3, 1, -2)$

B. $(2, 4, 4)$

C. $(6, 2, 4)$

D. $(-2, 1, 4)$

Question 8 [1 mark]

Categories: K, C
Which statement correctly describes skew lines in $\mathbb{R}^3$?

A. They are parallel and distinct.

B. They intersect at a right angle.

C. They are non-parallel and do not intersect.

D. They lie in the same plane.

Section B: Application

Question 9 [4 marks]

Categories: A, C
The line L passes through $A(-1, 2, 0)$ and $B(3, 0, 6)$. (a) Find a simplified direction vector. (b) Write vector and symmetric equations. (c) Verify that B satisfies your vector equation.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: A, T
Determine whether $L_1 : \vec{r} = (1, 0, 1) + t(1, 2, -1)$ and $L_2 : \vec{r} = (3, 5, -2) + s(0, 1, -1)$ intersect. Find the intersection if it exists and verify all three coordinates.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 11 [3 marks]

Categories: T, C
Explain how you would test whether $S(7, -2, 5)$ lies on $\vec{r} = (1, 1, 2) + t(2, -1, 1)$. State the conclusion.
Show your working and state a complete conclusion.

Question 12 [4 marks]

Categories: T, C
A student claims: ”If two line directions have dot product zero, the lines must intersect at a right angle.” Use $L_1 : \vec{r} = t(1, 0, 0)$ and $L_2 : \vec{r} = (0, 1, 1) + s(0, 1, 0)$ to assess this claim.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: B

$(3, -2, 5)$

The coefficient of the parameter is the direction vector. Any non-zero scalar multiple would also be valid.

1 Match the template. Compare with r=r₀+td.

$$\vec{r}_0 = (2, -1, 4), \qquad \vec{d} = (3, -2, 5)$$

Question 2 solution

Correct answer: B

$3x - 4y - 10 = 0$

A normal perpendicular to (4,3) is (3,-4). Using the point gives 3(x-2)-4(y+1)=0.

1 Normal. Choose n=(3,-4), perpendicular to d.

$$(3, -4) \cdot (4, 3) = 12 - 12 = 0$$

2 Point-normal form. Use the given point.

$$3(x - 2) - 4(y + 1) = 0 \Rightarrow 3x - 4y - 10 = 0$$

Question 3 solution

Correct answer: B

$\frac{x-1}{2} = \frac{z-4}{-1},\ y = 3$

A zero y-direction means y never changes. State y=3 separately; never divide by zero.

1 Read coordinates. The parametric system is x=1+2t, y=3, z=4-t.

$$t = \frac{x - 1}{2} = \frac{z - 4}{-1}, \qquad y = 3$$

Question 4 solution

Correct answer: B

The same line

The second direction is twice the first, and its anchor is the point obtained from the first line when t=2.

1 Compare directions. The direction vectors are parallel.

$$(4, -2, 6) = 2(2, -1, 3)$$

2 Test the second anchor. Use t=2 on the first line.

$$(1, 0, 2) + 2(2, -1, 3) = (5, -2, 8)$$

Question 5 solution

Correct answer: A

One linear equation in R³ defines a plane.

One independent linear constraint reduces three-dimensional space to a two-dimensional plane. A Cartesian line needs two independent plane equations.

1 Count dimensions. One independent constraint removes one degree of freedom.

$$3 - 1 = 2 \quad \text{dimensions}$$

Question 6 solution

Correct answer: B

7

Square all three components before adding; the magnitude is non-negative.

1 Reasoning. Square all three components before adding; the magnitude is non-negative.

$$\|\vec{d}\| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{49} = 7$$

Question 7 solution

Correct answer: A

$(3, 1, -2)$

Subtract the points, then simplify by a non-zero common factor.

1 Reasoning. Subtract the points, then simplify by a non-zero common factor.

$$\overrightarrow{AB} = (6, 2, -4) = 2(3, 1, -2)$$

Question 8 solution

Correct answer: C

They are non-parallel and do not intersect.

Skew lines are non-coplanar. Both non-parallelism and failure to intersect must be established.

1 Reasoning. Skew lines are non-coplanar. Both non-parallelism and failure to intersect must be established.

$$\vec{d_1} \times \vec{d_2} \neq \vec{0}, \qquad L_1 \cap L_2 = \varnothing$$

Question 9 solution

Model response

A suitable direction is $(2, -1, 3)$. Thus $\vec{r} = (-1, 2, 0) + t(2, -1, 3)$, $t \in \mathbb{R}$, and $\frac{x+1}{2} = \frac{y-2}{-1} = \frac{z}{3}$. At $t = 2$, the vector equation gives $(3, 0, 6) = B$.

Marking criteria

1. Finds the displacement and simplifies it correctly (1 mark).
2. Writes a vector equation with one shared parameter (1 mark).
3. Writes a valid symmetric equation (1 mark).
4. Verifies B with the same parameter in every coordinate (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Direction. Subtract A from B and divide by 2.

$$\overrightarrow{AB} = (4, -2, 6) = 2(2, -1, 3)$$

2 Represent. Use A as the anchor.

$$\vec{r} = (-1, 2, 0) + t(2, -1, 3), \qquad \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z}{3}$$

3 Verify. Use one parameter value for all coordinates.

$$\vec{r}(2) = (-1, 2, 0) + 2(2, -1, 3) = (3, 0, 6)$$

Question 10 solution

Model response

From the x-coordinate, $1 + t = 3$, so $t = 2$. From y, $2t = 5 + s$, so $s = -1$. The z-coordinates agree: $1 - 2 = -2 - (-1) = -1$. Therefore the lines intersect at $(3, 4, -1)$.

Marking criteria

1. Equates corresponding coordinates using distinct parameters (1 mark).
2. Finds t=2 and s=-1 (1 mark).
3. Checks the remaining coordinate (1 mark).
4. States the intersection point clearly (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Solve. Equate x and y.

$$1 + t = 3 \Rightarrow t = 2, \qquad 2t = 5 + s \Rightarrow s = -1$$

2 Check z. The third equation must also hold.

$$1 - t = -1 = -2 - s$$

3 Conclude. Evaluate either line.

$$L_1 \cap L_2 = \{(3, 4, -1)\}$$

Question 11 solution

Model response

From x, $7 = 1 + 2t$ gives $t = 3$. Then $y = 1 - 3 = -2$ and $z = 2 + 3 = 5$. The same value $t = 3$ satisfies every coordinate, so S lies on the line.

Marking criteria

1. Solves for a parameter using at least one coordinate.
2. Checks the same parameter in the remaining coordinates.
3. States a clear geometric conclusion.

A point lies on a parametric line only when one shared parameter reproduces every coordinate.

1 Use x. Solve for the shared parameter.

$$7 = 1 + 2t \Rightarrow t = 3$$

2 Check y and z. Use the same value of t.

$$1 - 3 = -2, \qquad 2 + 3 = 5$$

3 Conclude. All coordinates agree.

$$S \in L$$

Question 12 solution

Model response

The directions are orthogonal because $(1, 0, 0) \cdot (0, 1, 0) = 0$. However, L_1 has $z = 0$ and L_2 has $z = 1$, so they cannot meet. They are non-parallel and disjoint, hence skew. To conclude that two lines meet at a right angle, an intersection must also be shown.

Marking criteria

1. Computes the direction dot product as zero (1 mark).
2. Observes that the first line has z=0 and the second has z=1 (1 mark).
3. Classifies the lines as skew (1 mark).
4. Explains that orthogonal directions alone do not establish intersection (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Direction test. A zero dot product concerns directions, not position.

$$(1,0,0)\cdot(0,1,0)=0$$

2 Position test. The z-coordinate equations are incompatible.

$$z_{L_1}=0,\quad z_{L_2}=1\Rightarrow L_1\cap L_2=\varnothing$$

4.2 Plane construction

Plane equations, independent spans, normal vectors, and verification.

12 questions 23 marks Suggested time: 30 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: K

What is a normal vector to $2x - y + 3z = 7$?

A. $(2, -1, 3)$

B. $(2, 1, 3)$

C. $(7, 0, 0)$

D. $(-1, 3, 7)$

Question 2 [1 mark]

Categories: A

A plane passes through P(1,0,2) and has spans a=(1,1,0), b=(0,1,1). Which scalar equation is correct?

A. $x + y + z = 3$

B. $x - y + z = 3$

C. $x - y - z = -1$

D. $x + z = 3$

Question 3 [1 mark]

Categories: T, C

Why do three collinear points fail to determine a unique plane?

A. Their two connecting vectors are perpendicular.

B. Their two connecting vectors are scalar multiples, so no unique normal is produced.

C. Every three points always determine a unique plane.

D. A plane cannot contain a line.

Question 4 [1 mark]

Categories: K, A
Which vector equation describes the plane through $P(1,0,2)$ with spans $(1,1,0)$ and $(0,1,1)$?

A. $\vec{r} = (1,0,2) + s(1,1,0) + t(0,1,1)$

B. $\vec{r} = (1,1,0) + t(0,1,1)$

C. $\vec{r} = (1,0,2) + s(1,1,0)$

D. $\vec{r} = s(1,0,2) + t(1,1,0)$

Question 5 [1 mark]

Categories: A
What are the axis intercepts of $2x + 3y + 6z = 6$?

A. $(3,0,0), (0,2,0), (0,0,1)$

B. $(2,0,0), (0,3,0), (0,0,6)$

C. $(6,0,0), (0,6,0), (0,0,6)$

D. $(1,0,0), (0,2,0), (0,0,3)$

Question 6 [1 mark]

Categories: A
Which point lies on $\vec{r} = (1,0,2) + s(1,1,0) + t(0,1,1)$?

A. $(3,3,3)$

B. $(3,2,3)$

C. $(0,0,0)$

D. $(1,1,2)$

Question 7 [1 mark]

Categories: A
A plane has spans $\vec{a} = (1,0,2)$ and $\vec{b} = (0,1,-1)$. Which vector is normal to the plane?

A. $(1,1,1)$

B. $(-2,1,1)$

C. $(2,1,1)$

D. $(1,0,2)$

Question 8 [1 mark]

Categories: K, T
Which plane is parallel to, but distinct from, $2x - y + z = 3$?

A. $4x - 2y + 2z = 6$

B. $4x - 2y + 2z = 5$

C. $x + y + z = 3$

D. $2x + y - z = 3$

Section B: Application

Question 9 [4 marks]

Categories: A, T
Find a Cartesian equation of the plane through $A(1,1,0)$, $B(2,0,1)$, and $C(0,2,1)$. Show that the points are not collinear and verify your equation.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: A, C
For the plane $2x + y - z = 4$, (a) find an anchor point and two independent span vectors, and (b) write and verify a vector equation.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 11 [3 marks]

Categories: T, C
Explain why two parallel span vectors cannot define a unique plane. Include the geometric meaning of their cross product.
Show your working and state a complete conclusion.

Question 12 [4 marks]

Categories: T, C
A proposed plane is $\vec{r} = (1,0,0) + s(1,2,-1) + t(k,4,-2)$. Find the value of k for which this expression fails to describe a plane. Explain what set the expression describes in that case.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: A

$(2, -1, 3)$

The coefficients of x, y, and z form a normal vector.

1 Read coefficients. Use A, B, C from Ax+By+Cz=D.

$$\vec{n} = (A, B, C) = (2, -1, 3)$$

Question 2 solution

Correct answer: B

$x - y + z = 3$

a×b=(1,-1,1), and the point-normal equation expands to x-y+z=3.

1 Normal. Cross the two spans.

$$(1, 1, 0) \times (0, 1, 1) = (1, -1, 1)$$

2 Equation. Use P.

$$(x - 1) - y + (z - 2) = 0 \Rightarrow x - y + z = 3$$

Question 3 solution

Correct answer: B

Their two connecting vectors are scalar multiples, so no unique normal is produced.

Collinearity makes the two candidate span vectors dependent; their cross product is zero.

1 Independence test. For collinear A, B, C, AC=kAB.

$$\overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$$

Question 4 solution

Correct answer: A

$\vec{r} = (1, 0, 2) + s(1, 1, 0) + t(0, 1, 1)$

A plane needs one anchor and two independent spanning directions.

1 Use the plane template. Insert the anchor and both spans.

$$\vec{r} = \vec{r_0} + s\vec{a} + t\vec{b}$$

Question 5 solution

Correct answer: A

$(3,0,0),(0,2,0),(0,0,1)$

Set the other two coordinates to zero for each intercept.

1 Calculate each intercept. Solve one variable at a time.

$$x=3, \qquad y=2, \qquad z=1$$

Question 6 solution

Correct answer: A

$(3,3,3)$

The coordinates require x=1+s, y=s+t, and z=2+t. With s=2 and t=1, all three coordinates give the first point.

1 Reasoning. The coordinates require x=1+s, y=s+t, and z=2+t. With s=2 and t=1, all three coordinates give the first point.

$$(1,0,2)+2(1,1,0)+(0,1,1)=(3,3,3)$$

Question 7 solution

Correct answer: B

$(-2,1,1)$

The cross product is perpendicular to both independent spans.

1 Reasoning. The cross product is perpendicular to both independent spans.

$$(1,0,2)\times(0,1,-1)=(-2,1,1)$$

Question 8 solution

Correct answer: B

$4x-2y+2z=5$

The normal scales by 2 while the constant does not. A constant of 6 would give the same plane.

1 Reasoning. The normal scales by 2 while the constant does not. A constant of 6 would give the same plane.

$$2(2x-y+z)=4x-2y+2z, \qquad 2(3)=6\neq 5$$

Question 9 solution

Model response

The spans are $\overrightarrow{AB} = (1, -1, 1)$ and $\overrightarrow{AC} = (-1, 1, 1)$. Their cross product is $(-2, -2, 0) \neq \vec{0}$, so the points are not collinear. Use the normal $(1, 1, 0)$ and A to obtain $x + y = 2$. Each given point has coordinate sum $x + y = 2$.

Marking criteria

1. Finds two correct connecting vectors (1 mark).
2. Finds a non-zero cross product and explains non-collinearity (1 mark).
3. Obtains x+y=2 (1 mark).
4. Checks all three points (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Find spans. Use a common anchor.

$$\overrightarrow{AB} = (1, -1, 1), \qquad \overrightarrow{AC} = (-1, 1, 1)$$

2 Find normal. A non-zero cross product also confirms independence.

$$\overrightarrow{AB} \times \overrightarrow{AC} = (-2, -2, 0)$$

3 Equation and check. Scale the normal and use A.

$$(x - 1) + (y - 1) = 0 \Rightarrow x + y = 2$$

Question 10 solution

Model response

Choose $A = (2, 0, 0)$, $\vec{a} = (1, -2, 0)$, and $\vec{b} = (0, 1, 1)$. Both spans are perpendicular to $(2, 1, -1)$ and they are not scalar multiples. A vector equation is $\vec{r} = (2, 0, 0) + s(1, -2, 0) + t(0, 1, 1)$. Substituting $x = 2 + s$, $y = -2s + t$, $z = t$ gives 4 for all parameters.

Marking criteria

1. Finds an anchor satisfying the scalar equation (1 mark).
2. Finds two in-plane directions perpendicular to the normal (1 mark).
3. Establishes that the directions are independent (1 mark).
4. Writes and verifies a valid vector equation (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Anchor. Set y=z=0.

$$2x = 4 \Rightarrow A = (2, 0, 0)$$

2 Directions. Choose two independent solutions of 2a+b-c=0.

$$(2,1,-1)\cdot(1,-2,0)=0, \quad (2,1,-1)\cdot(0,1,1)=0$$

3 Verify. Substitute the full parametric representation.

$$2(2+s)+(-2s+t)-t=4$$

Question 11 solution

Model response

Parallel span vectors are scalar multiples, so they describe movement in only one independent direction. Their cross product is the zero vector, which cannot serve as a plane normal. Therefore the data determines a line of movement but not one unique plane.

Marking criteria

1. States that parallel spans provide only one independent direction.
2. Identifies the cross product as the zero vector.
3. Explains why no unique normal or plane is produced.

A plane is two-dimensional and therefore requires two independent directions.

1 Identify dependence. Parallel vectors are scalar multiples.

$$\vec{b}=k\vec{a}$$

2 Use the cross product. Dependent spans produce no non-zero normal.

$$\vec{a}\times\vec{b}=\vec{0}$$

Question 12 solution

Model response

For dependence, $(k,4,-2)=\lambda(1,2,-1)$. The y- and z-components give $\lambda=2$, so $k=2$. Then $\vec{r}=(1,0,0)+(s+2t)(1,2,-1)$, which generates a line, not a plane. For every other k, the spans are independent.

Marking criteria

1. Uses scalar dependence of the two spans (1 mark).
2. Finds the common scale factor 2 (1 mark).
3. Obtains k=2 (1 mark).
4. Explains that only a line is generated (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Compare spans. Use the non-zero components to determine the scale factor.

$$4 = 2\lambda, \quad -2 = -\lambda \Rightarrow \lambda = 2$$

2 Interpret. At k=2 the two parameters combine into one effective parameter.

$$\vec{r} = (1, 0, 0) + (s + 2t)(1, 2, -1)$$

4.3 Intersections and distance

Intersections, exact distances, angles, and parameter-dependent relationships.

12 questions 23 marks Suggested time: 35 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: A
Where does $\vec{r} = (1, 0, 2) + t(1, 2, -1)$ meet $x + y + z = 5$?

A. $(1, 0, 2)$

B. $(2, 2, 1)$

C. $(3, 4, 0)$

D. It never meets the plane

Question 2 [1 mark]

Categories: T, C
If $\vec{n} \cdot \vec{d} = 0$ for a line and plane, but the line anchor does not satisfy the plane, what is the relationship?

A. One intersection

B. The line lies in the plane

C. The line is parallel and disjoint

D. The line is perpendicular to the plane

Question 3 [1 mark]

Categories: A, T
Find the distance from $P(1, 0, 2)$ to $\vec{r} = t(1, 1, 0)$.

A. $\sqrt{2}$

B. $3/\sqrt{2}$

C. $\sqrt{5}$

D. $1/\sqrt{2}$

Question 4 [1 mark]

Categories: K
Which expression gives the distance from point P to a line through A with direction d in R³?

A. $\frac{\|\overrightarrow{AP}\times\vec{d}\|}{\|\vec{d}\|}$

B. $\overrightarrow{AP}\cdot\vec{d}$

C. $\|\overrightarrow{AP}+\vec{d}\|$

D. $\frac{\|\vec{d}\|}{\|\overrightarrow{AP}\times\vec{d}\|}$

Question 5 [1 mark]

Categories: A, T
What direction vector is parallel to the intersection of $x+y+z=3$ and $x-y+z=1$?

A. $(-1,0,1)$

B. $(1,1,1)$

C. $(1,-1,1)$

D. $(0,1,0)$

Question 6 [1 mark]

Categories: A
Find the distance from the origin to $2x-y+2z=6$.

A. 2

B. 3

C. 6

D. $\sqrt{6}$

Question 7 [1 mark]

Categories: K, A
A line has direction $(1,0,1)$. What is its acute angle with the plane $z=0$?

A. $0°$

B. $30°$

C. $45°$

D. $90°$

Question 8 [1 mark]

Categories: T
A system includes $x+y+z=1$ and $2x+2y+2z=3$. What can be concluded before examining a third plane equation?

A. There is exactly one solution.

B. There are infinitely many solutions.

C. There is no common solution.

D. The common solution is the origin.

Section B: Application

Question 9 [4 marks]

Categories: A, C
Find a vector equation for the intersection of $x + y + z = 4$ and $x - y + z = 2$. Verify both an anchor and the direction vector.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: A, T
For $P(2, 1, 3)$ and the plane $x + 2y + 2z = 1$, (a) calculate the exact perpendicular distance and (b) find and verify the foot of the perpendicular.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 11 [3 marks]

Categories: T, C
A line has direction $\vec{d} = (1, -1, 0)$, anchor $A = (0, 0, 0)$, and the plane is $x + y + z = 2$. Classify the relationship and justify it without solving a full substitution system.
Show your working and state a complete conclusion.

Question 12 [4 marks]

Categories: T, C
Consider $L : \vec{r} = (1, 2, 0) + t(1, 1, -1)$ and $\Pi_k : 2x - y + z = k$. Determine all values of k for which the line (a) lies in the plane or (b) is parallel and disjoint. Can there be exactly one intersection?
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: B

$(2, 2, 1)$

Substitution gives 3+2t=5, so t=1 and the point is (2,2,1).

1 Substitute. Use x=1+t, y=2t, z=2-t.

$$1 + t + 2t + 2 - t = 5 \Rightarrow t = 1$$

2 Evaluate. Return to the line.

$$\vec{r}(1) = (2, 2, 1)$$

Question 2 solution

Correct answer: C

The line is parallel and disjoint

n·d=0 makes the line direction parallel to the plane. Since its anchor is outside, the entire line is disjoint.

1 Classify. The substitution reduces to a contradiction rather than 0=0.

$$\vec{n} \cdot \vec{d} = 0, \quad \vec{r}_0 \notin \Pi \Rightarrow L \parallel \Pi$$

Question 3 solution

Correct answer: B

$3/\sqrt{2}$

AP×d=(-2,2,1), with magnitude 3; ||d||=√2.

1 Cross product. The line anchor is the origin, so AP=P.

$$(1, 0, 2) \times (1, 1, 0) = (-2, 2, 1)$$

2 Distance. Divide magnitudes.

$$d = \frac{\sqrt{4 + 4 + 1}}{\sqrt{1 + 1}} = \frac{3}{\sqrt{2}}$$

Question 4 solution

Correct answer: A

$\frac{\|\overrightarrow{AP}\times\vec{d}\|}{\|\vec{d}\|}$

The cross-product magnitude measures the parallelogram area. Dividing by the base length leaves the perpendicular height.

1 Area relation. Area equals base times height.

$$\|\overrightarrow{AP}\times\vec{d}\| = \|\vec{d}\|d$$

Question 5 solution

Correct answer: A

$(-1,0,1)$

The intersection direction is perpendicular to both plane normals, so it is parallel to their cross product.

1 Cross the normals. Use n_1=(1,1,1) and n_2=(1,-1,1).

$$\vec{n}_1\times\vec{n}_2 = (2,0,-2) \parallel (-1,0,1)$$

Question 6 solution

Correct answer: A

2

Divide the absolute substitution residual by the normal magnitude.

1 Reasoning. Divide the absolute substitution residual by the normal magnitude.

$$d = \frac{|0-6|}{\sqrt{2^2+(-1)^2+2^2}} = \frac{6}{3} = 2$$

Question 7 solution

Correct answer: C

45°

Use the sine of the line-plane angle with normal (0,0,1).

1 Reasoning. Use the sine of the line-plane angle with normal (0,0,1).

$$\sin\theta = \frac{|(1,0,1)\cdot(0,0,1)|}{\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

Question 8 solution

Correct answer: C

There is no common solution.

The first two equations already contradict one another; a third equation cannot restore consistency.

1 Reasoning. The first two equations already contradict one another; a third equation cannot restore consistency.

$$(2x + 2y + 2z) - 2(x + y + z) = 3 - 2 \Rightarrow 0 = 1$$

Question 9 solution

Model response

Subtracting gives $2y = 2$, so $y = 1$. Then $x + z = 3$. Set $z = t$ to obtain $\vec{r} = (3, 1, 0) + t(-1, 0, 1)$. The anchor satisfies both equations, and $(-1, 0, 1)$ has zero dot product with both normals.

Marking criteria

1. Eliminates a variable correctly (1 mark).
2. Finds an anchor on both planes (1 mark).
3. Finds a direction and vector equation (1 mark).
4. Verifies the anchor and direction against both planes (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Eliminate. Subtract the second plane from the first.

$$2y = 2 \Rightarrow y = 1, \qquad x + z = 3$$

2 Parameterize. Choose z as the free parameter.

$$\vec{r} = (3, 1, 0) + t(-1, 0, 1)$$

3 Verify. The direction is perpendicular to each normal.

$$(1, 1, 1) \cdot (-1, 0, 1) = 0, \quad (1, -1, 1) \cdot (-1, 0, 1) = 0$$

Question 10 solution

Model response

The normal is $\vec{n} = (1, 2, 2)$, with magnitude 3. The distance is $|2 + 2 + 6 - 1|/3 = 3$. The foot is $Q = P - \frac{9}{9}\vec{n} = (1, -1, 1)$. It satisfies $1 - 2 + 2 = 1$, and $P - Q = (1, 2, 2)$ is normal to the plane.

Marking criteria

1. Uses the point-plane formula correctly (1 mark).
2. Obtains distance 3 (1 mark).
3. Finds the perpendicular foot (1,-1,1) (1 mark).
4. Verifies plane membership and perpendicular displacement (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Distance. Use the residual and normal magnitude.

$$d = \frac{|2 + 2(1) + 2(3) - 1|}{\sqrt{1 + 4 + 4}} = 3$$

2 Foot. Remove the normal component of the displacement.

$$Q = P - \frac{\vec{n} \cdot P - 1}{\|\vec{n}\|^2}\vec{n} = (2, 1, 3) - (1, 2, 2) = (1, -1, 1)$$

3 Verify. Check membership and perpendicular direction.

$$1 + 2(-1) + 2(1) = 1, \qquad P - Q = \vec{n}$$

Question 11 solution

Model response

The plane normal is n=(1,1,1). Since n dot d=1-1+0=0, the line direction is parallel to the plane. The anchor does not satisfy the plane because 0+0+0 is not 2. Therefore the line is parallel to the plane and has no intersection.

Marking criteria

1. Uses the plane normal n=(1,1,1).
2. Computes n dot d=0 and explains that the line direction is parallel to the plane.
3. Tests the anchor and concludes that the line is parallel and disjoint.

The dot product identifies parallel direction, while the anchor test distinguishes a contained line from a disjoint line.

1 Direction test. Compare the normal and line direction.

$$(1, 1, 1) \cdot (1, -1, 0) = 0$$

2 Anchor test. Substitute A into the plane.

$$0 + 0 + 0 \neq 2$$

Question 12 solution

Model response

Since $(2,-1,1)\cdot(1,1,-1)=0$, the line direction is parallel to every plane in the family. Substitution gives $2(1+t)-(2+t)-t=0$. Thus the line lies in Π_0, is parallel and disjoint when $k\neq 0$, and never has exactly one intersection.

Marking criteria

1. Calculates the normal-direction dot product as zero (1 mark).
2. Obtains the constant substitution value 0 (1 mark).
3. Distinguishes k=0 from k not equal to 0 (1 mark).
4. Explains why exactly one intersection is impossible (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Direction. Use the same normal for the plane family.

$$(2,-1,1)\cdot(1,1,-1)=2-1-1=0$$

2 Substitute. The parameter cancels.

$$2(1+t)-(2+t)-t=0$$

3 Classify. Compare the constant expression with k.

$$k=0\Rightarrow L\subset\Pi_k,\qquad k\neq 0\Rightarrow L\cap\Pi_k=\varnothing$$

5 Unit tests

5.1 Lines and plane construction

A 40-mark unit test on line equations, plane construction, vector reasoning, and verification.

16 questions 40 marks Suggested time: 60 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: K
Read a direction vector from $x = 3 - 2t,\ y = 4,\ z = 1 + 5t$.

A. $(3, 4, 1)$

B. $(-2, 0, 5)$

C. $(-2, 4, 5)$

D. $(2, 0, 5)$

Question 2 [1 mark]

Categories: K
Which vector is normal to the line $5x + 2y - 7 = 0$ in $\mathbb{R}^2$?

A. $(2, -5)$

B. $(5, -2)$

C. $(5, 2)$

D. $(7, 0)$

Question 3 [1 mark]

Categories: A
Which point does NOT lie on $\vec{r} = (0, 1, 2) + t(1, -1, 2)$?

A. $(1, 0, 4)$

B. $(2, -1, 6)$

C. $(-1, 2, 0)$

D. $(1, 0, 3)$

Question 4 [1 mark]

Categories: K, A
Which plane passes through the origin and has normal $(2, -1, 3)$?

A. $2x - y + 3z = 0$

B. $x + 2y + 3z = 0$

C. $2x - y + 3z = 1$

D. $2x + y - 3z = 0$

Question 5 [1 mark]

Categories: A, T
Which pair can serve as independent span vectors for $x + 2y + z = 0$?

A. $(2, -1, 0), (1, 0, -1)$

B. $(1, 2, 1), (2, 4, 2)$

C. $(1, 0, 0), (0, 1, 0)$

D. $(2, -1, 0), (4, -2, 0)$

Question 6 [1 mark]

Categories: K, A
What is the relationship between $x + 2y + 2z = 1$ and $2x + y - 2z = 3$?

A. Coincident

B. Parallel and distinct

C. Perpendicular

D. They meet at exactly one point

Question 7 [1 mark]

Categories: A
Find the vector projection of $\vec{u} = (2, 1, 2)$ onto $\vec{d} = (1, 0, 1)$.

A. $(1, 0, 1)$

B. $(2, 0, 2)$

C. $(0, 1, 0)$

D. $(4, 0, 4)$

Question 8 [1 mark]

Categories: K, T
How many planes contain all three points $(1, 2, 3)$, $(2, 4, 6)$, and $(3, 6, 9)$?

A. None

B. Exactly one

C. Exactly two

D. Infinitely many

Section B: Application

Question 9 [4 marks]

Categories: A, C
A line passes through $A(2,-1,3)$ and $B(-2,5,1)$. (a) Find a simplified direction and its exact magnitude. (b) Write vector and symmetric equations.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: A, C
In $\mathbb{R}^2$, a line passes through $A(-2,3)$ with direction $(3,-4)$. (a) Find a scalar equation. (b) Decide whether $P(1,-1)$ lies on it, showing a check.
Show your working and state a complete conclusion.

Question 11 [4 marks]

Categories: A, T, C
Find a Cartesian equation of the plane through $A(1,0,0)$, $B(0,2,0)$, and $C(0,0,3)$. Verify all three points and explain why the equation is unique up to a non-zero scalar multiple.
Show your working and state a complete conclusion.

Question 12 [4 marks]

Categories: A, T
Construct a Cartesian equation of the plane containing $L:\vec{r}=(1,-1,2)+t(2,1,0)$ and $Q(0,1,3)$. Verify both the point Q and the entire line.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 13 [4 marks]

Categories: T, C
Classify $L_1:\vec{r}=(0,1,0)+t(1,0,1)$ and $L_2:\vec{r}=(2,1,3)+s(2,0,2)$. Explain why comparing the direction vectors alone is insufficient.
Show your working and state a complete conclusion.

Question 14 [4 marks]

Categories: T, C
A student uses $(3,-2,1)$ as an anchor for the plane $3x-2y+z=6$ because these are its coefficients. Explain the error, test that proposed anchor, and provide a correct anchor with a point-normal equation.
Show your working and state a complete conclusion.

Question 15 [4 marks]

Categories: T, C
For which real values of k does $\vec{r}=s(1,2,3)+t(2,4,k)$ describe a plane? Show a vector test and describe the exceptional case.
Show your working and state a complete conclusion.

Question 16 [4 marks]

Categories: T, C
The lines $L_1 : \vec{r} = (1, 0, 0) + t(1, 1, 0)$ and $L_2 : \vec{r} = (0, 0, 1) + s(1, 1, 0)$ are parallel and distinct. Find the plane containing them. Explain why the common direction alone does not determine this plane.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: B

$(-2, 0, 5)$

The coefficients of the shared parameter are the direction components. The constant y-coordinate has direction component zero.

1 Reasoning. The coefficients of the shared parameter are the direction components. The constant y-coordinate has direction component zero.

$$\vec{d} = (-2, 0, 5)$$

Question 2 solution

Correct answer: C

$(5, 2)$

For Ax+By+C=0, the coefficient vector (A,B) is a normal.

1 Reasoning. For Ax+By+C=0, the coefficient vector (A,B) is a normal.

$$\vec{n} = (5, 2)$$

Question 3 solution

Correct answer: D

$(1, 0, 3)$

For the final point, x=1 requires t=1, which gives z=4 rather than z=3.

1 Reasoning. For the final point, x=1 requires t=1, which gives z=4 rather than z=3.

$$t = 1 \Rightarrow \vec{r} = (1, 0, 4) \neq (1, 0, 3)$$

Question 4 solution

Correct answer: A

$2x - y + 3z = 0$

Use the normal as the coefficient vector. The constant must be zero because the origin lies on the plane.

1 Reasoning. Use the normal as the coefficient vector. The constant must be zero because the origin lies on the plane.

$$(2, -1, 3) \cdot (x, y, z) = 0$$

Question 5 solution

Correct answer: A

$(2,-1,0),(1,0,-1)$

Both listed spans must be perpendicular to the normal and must not be parallel to one another.

1 Reasoning. Both listed spans must be perpendicular to the normal and must not be parallel to one another.

$$(1,2,1)\cdot(2,-1,0)=0,\quad (1,2,1)\cdot(1,0,-1)=0$$

Question 6 solution

Correct answer: C

Perpendicular

Their non-zero normals have dot product zero, so the planes are perpendicular.

1 Reasoning. Their non-zero normals have dot product zero, so the planes are perpendicular.

$$(1,2,2)\cdot(2,1,-2)=2+2-4=0$$

Question 7 solution

Correct answer: B

$(2,0,2)$

Multiply the target direction by the dot product divided by its squared magnitude.

1 Reasoning. Multiply the target direction by the dot product divided by its squared magnitude.

$$\text{proj}_{\vec{d}}\,\vec{u}=\frac{4}{2}(1,0,1)=(2,0,2)$$

Question 8 solution

Correct answer: D

Infinitely many

The points are collinear. Infinitely many planes contain their common line.

1 Reasoning. The points are collinear. Infinitely many planes contain their common line.

$$\overrightarrow{AB}=(1,2,3),\quad \overrightarrow{AC}=2(1,2,3)$$

Question 9 solution

Model response

Since $B - A = (-4, 6, -2) = 2(-2, 3, -1)$, choose $\vec{d} = (-2, 3, -1)$, with $\|\vec{d}\| = \sqrt{14}$. Then $\vec{r} = (2, -1, 3) + t(-2, 3, -1)$ and $\frac{x-2}{-2} = \frac{y+1}{3} = \frac{z-3}{-1}$.

Marking criteria

1. Finds a simplified direction (-2,3,-1) (1 mark).
2. Gives exact magnitude square root of 14 (1 mark).
3. Writes a valid vector equation (1 mark).
4. Writes a valid symmetric equation (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Direction and magnitude. Simplify the displacement before calculating its magnitude.

$$\vec{d} = (-2, 3, -1), \qquad \|\vec{d}\| = \sqrt{4 + 9 + 1} = \sqrt{14}$$

2 Represent. Use A as the anchor.

$$\vec{r} = (2, -1, 3) + t(-2, 3, -1), \qquad \frac{x-2}{-2} = \frac{y+1}{3} = \frac{z-3}{-1}$$

Question 10 solution

Model response

A normal is $(4, 3)$, since $4(3) + 3(-4) = 0$. Point-normal form gives $4(x + 2) + 3(y - 3) = 0$, so $4x + 3y - 1 = 0$. For P, $4(1) + 3(-1) - 1 = 0$; hence P lies on the line.

Marking criteria

1. Finds a perpendicular normal (4,3) (1 mark).
2. Uses the given point correctly (1 mark).
3. Obtains 4x+3y-1=0 (1 mark).
4. Verifies that P lies on the line (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Normal. Choose a perpendicular vector.

$$(4, 3) \cdot (3, -4) = 0$$

2 Equation. Insert A into point-normal form.

$$4(x + 2) + 3(y - 3) = 0 \Rightarrow 4x + 3y - 1 = 0$$

3 Membership. Substitute both coordinates of P.

$$4(1) + 3(-1) - 1 = 0$$

Question 11 solution

Model response

The connecting vectors $(-1, 2, 0)$ and $(-1, 0, 3)$ have cross product $(6, 3, 2)$. Using A gives $6x + 3y + 2z = 6$. Substitution gives 6 at each point. The cross product is non-zero, so the points are not collinear and determine one plane. Multiplying the whole equation by a non-zero number changes only its representation.

Marking criteria

1. Finds two connecting vectors and a non-zero normal (1 mark).
2. Obtains 6x+3y+2z=6 (1 mark).
3. Checks all three points (1 mark).
4. Uses non-collinearity to justify a unique plane (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Normal. Cross the connecting vectors.

$$(-1, 2, 0) \times (-1, 0, 3) = (6, 3, 2) \neq \vec{0}$$

2 Equation. Use A in point-normal form.

$$6(x - 1) + 3y + 2z = 0$$

3 Verify. Each intercept satisfies the equation.

$$6(1) = 3(2) = 2(3) = 6$$

Question 12 solution

Model response

From the anchor A to Q, $\overrightarrow{AQ} = (-1, 2, 1)$. A normal is $(2, 1, 0) \times (-1, 2, 1) = (1, -2, 5)$. Thus $x - 2y + 5z = 13$. Q gives $0 - 2 + 15 = 13$. For the line, $(1 + 2t) - 2(-1 + t) + 5(2) = 13$ for every t.

Marking criteria

1. Forms a connector from the line anchor to Q (1 mark).
2. Finds a valid non-zero normal (1 mark).
3. Obtains x-2y+5z=13 (1 mark).
4. Checks Q and the line parametrically (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Normal. Use the line direction and a connector to Q.

$$(2,1,0)\times(-1,2,1)=(1,-2,5)$$

2 Equation. Use the line anchor.

$$(x-1)-2(y+1)+5(z-2)=0\Rightarrow x-2y+5z=13$$

3 Verify. The parameter cancels when the line is substituted.

$$(1+2t)-2(-1+t)+10=13$$

Question 13 solution

Model response

The second direction is twice the first, so the lines are parallel or coincident. The second anchor would require $t=2$ from x and $t=3$ from z on L_1, which is impossible. Thus the lines are parallel and distinct. An anchor test is needed to distinguish this from coincidence.

Marking criteria

1. Establishes parallel directions (1 mark).
2. Tests whether the second anchor lies on the first line (1 mark).
3. Shows that the coordinate conditions conflict (1 mark).
4. Concludes that the lines are parallel and distinct, not coincident (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Compare directions. A scalar multiple establishes parallelism.

$$(2,0,2)=2(1,0,1)$$

2 Test anchor. One shared parameter is required.

$$(t,1,t)=(2,1,3)\Rightarrow t=2 \text{ and } t=3$$

3 Conclusion. The anchors do not lie on the same line.

$$L_1\parallel L_2,\qquad L_1\neq L_2$$

Question 14 solution

Model response

The coefficient vector is a normal, not automatically a point on the plane. The proposed point gives $3(3)-2(-2)+1=14\neq 6$. A correct anchor is $(2,0,0)$. Point-normal form is $(3,-2,1)\cdot((x,y,z)-(2,0,0))=0$, or $3(x-2)-2y+z=0$.

Marking criteria

1. Distinguishes a normal vector from an anchor point (1 mark).
2. Shows that the proposed point gives 14, not 6 (1 mark).
3. Finds a correct anchor such as (2,0,0) (1 mark).
4. Writes correct point-normal form (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Test. Coordinates of an anchor must satisfy the scalar equation.

$$3(3) - 2(-2) + 1 = 14 \neq 6$$

2 Correct. Set y=z=0 and solve for x.

$$A = (2, 0, 0), \qquad 3(x - 2) - 2y + z = 0$$

Question 15 solution

Model response

The cross product is $(1, 2, 3) \times (2, 4, k) = (2k - 12, 6 - k, 0) = (k - 6)(2, -1, 0)$. This is non-zero exactly when $k \neq 6$, so those values give a plane. At $k = 6$, the second span is twice the first and the expression reduces to $\vec{r} = (s + 2t)(1, 2, 3)$, a line through the origin.

Marking criteria

1. Computes or correctly analyzes the cross product (1 mark).
2. Finds the exceptional value k=6 (1 mark).
3. States that all k not equal to 6 produce a plane (1 mark).
4. Explains that k=6 produces a line through the origin (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Independence. A plane requires a non-zero cross product.

$$(1, 2, 3) \times (2, 4, k) = (k - 6)(2, -1, 0)$$

2 Exceptional case. At k=6 only one independent direction remains.

$$\vec{r} = (s + 2t)(1, 2, 3)$$

Question 16 solution

Model response

Use the common direction $(1, 1, 0)$ and the connector $(-1, 0, 1)$. Their cross product is $(1, -1, 1)$. Using $(1, 0, 0)$ gives $x - y + z = 1$. Both anchors satisfy it, and the common direction is perpendicular to its normal. One direction alone leaves infinitely many planes; the connector supplies the second independent direction.

Marking criteria

1. Uses the connector (-1,0,1) as a second direction (1 mark).
2. Finds normal (1,-1,1) (1 mark).
3. Obtains x-y+z=1 (1 mark).
4. Explains the need for two independent in-plane directions (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Second span. Connect the two anchors.

$$(0, 0, 1) - (1, 0, 0) = (-1, 0, 1)$$

2 Normal and equation. Cross the independent spans.

$$(1, 1, 0) \times (-1, 0, 1) = (1, -1, 1), \qquad x - y + z = 1$$

5.2 Intersections, distances, and angles

A 40-mark unit test on spatial systems, exact measurement, and complete geometric justification.

16 questions 40 marks Suggested time: 65 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: K

A line has direction $(2, -1, 2)$. How is it oriented relative to $2x - y + 2z = 5$?

A. It is parallel to the plane.

B. It is perpendicular to the plane.

C. It must lie in the plane.

D. It is skew to the plane.

Question 2 [1 mark]

Categories: A

Find the distance from $(0, 0, 0)$ to $x - 2y + 2z = 9$.

A. 1

B. 3

C. 9

D. $\sqrt{5}$

Question 3 [1 mark]

Categories: K, T

Two planes have non-zero normals $\vec{n}_1, \vec{n}_2$ with $\vec{n}_1 \times \vec{n}_2 = \vec{0}$. What can be concluded from this test alone?

A. They are perpendicular.

B. They intersect in exactly one line.

C. They are parallel or coincident.

D. They have no common points.

Question 4 [1 mark]

Categories: A
Find the acute angle between $x = 0$ and $x + y = 0$.

A. $0°$

B. $30°$

C. $45°$

D. $90°$

Question 5 [1 mark]

Categories: K, T
A consistent three-variable system reduces to $x + y = 2$ and $y - z = 1$, with no other independent constraint. Its solution set is:

A. One point

B. A line

C. A plane

D. All of three-dimensional space

Question 6 [1 mark]

Categories: A
Find the distance from $P(4, 6, 5)$ to $\vec{r} = (1, 2, 0) + t(0, 0, 1)$.

A. $\sqrt{50}$

B. 5

C. $\sqrt{41}$

D. 9

Question 7 [1 mark]

Categories: A
Classify $\vec{r} = (1, 2, 0) + t(1, 0, 1)$ relative to $x + y - z = 3$.

A. Exactly one intersection

B. Contained in the plane

C. Parallel and disjoint

D. Perpendicular to the plane

Question 8 [1 mark]

Categories: T
For two non-parallel lines, let $\vec{c}$ connect their anchors. If $\vec{c} \cdot (\vec{d_1} \times \vec{d_2}) \neq 0$, the lines are:

A. Coincident

B. Intersecting

C. Parallel

D. Skew

Section B: Application

Question 9 [4 marks]

Categories: A, C
Find the intersection of $L : \vec{r} = (2, -1, 1) + t(1, 2, -1)$ and $\Pi : x - y + z = 0$. Verify that the point satisfies both representations.
Show your working and state a complete conclusion.

Question 10 [4 marks]

Categories: A, C
Determine the intersection line of $2x + y - z = 3$ and $x - y + z = 0$. Give vector form and verify your anchor and direction.
Show your working and state a complete conclusion.

Question 11 [4 marks]

Categories: A, T
Solve the three-plane system $x+y+z = 6$, $x-y+z = 2$, and $2x+y-z = 3$. State the geometric meaning and verify the result in all equations.
Show your working and state a complete conclusion.

Question 12 [4 marks]

Categories: A, T, C
Find the exact distance from $P(4, 1, 2)$ to $L : \vec{r} = (1, -1, 0) + t(1, 2, 2)$. Show the cross product used and explain why dividing by the direction magnitude is necessary.
Show your working and state a complete conclusion.

Question 13 [4 marks]

Categories: A, T, C
For $P(4, 1)$ and $L : \vec{r} = (-1, 2) + t(2, 1)$ in $\mathbb{R}^2$, use projection to find the closest point H on the line and the exact distance PH. Verify why this connector gives the shortest distance.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 14 [4 marks]

Categories: T, C
Find the acute angle between a line with direction $(2, 1, 2)$ and the plane $x + 2y + 2z = 0$. Give an exact inverse-trigonometric expression and an angle to the nearest tenth of a degree. Explain why sine, rather than cosine, appears.
Show your working and state a complete conclusion.

Question 15 [4 marks]

Categories: T, C
Classify the planes $x+y+z = 1$ and $2x+2y+kz = 3$ for all real values of k. For the intersecting case, find a vector equation of the common line.
Show your working and state a complete conclusion.

Question 16 [4 marks]

Categories: T, C
A student substitutes $L : \vec{r} = (1, 0, 1) + t(1, -1, 0)$ into $\Pi : x + y + z = 2$, obtains $0 = 0$, and concludes "one intersection at t=0." Correct the reasoning and state the complete intersection set. Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: B

It is perpendicular to the plane.

The line direction equals the plane normal, so the line is perpendicular to the plane.

1 Reasoning. The line direction equals the plane normal, so the line is perpendicular to the plane.

$$\vec{d} = (2, -1, 2) = \vec{n}$$

Question 2 solution

Correct answer: B

3

The normal has magnitude 3, while the absolute residual at the origin is 9.

1 Reasoning. The normal has magnitude 3, while the absolute residual at the origin is 9.

$$d = \frac{9}{\sqrt{1+4+4}} = 3$$

Question 3 solution

Correct answer: C

They are parallel or coincident.

Parallel normals imply parallel orientations. Constants or a shared-point test are still required to distinguish distinct planes from coincidence.

1 Reasoning. Parallel normals imply parallel orientations. Constants or a shared-point test are still required to distinguish distinct planes from coincidence.

$$\vec{n}_2 = \lambda \vec{n}_1 \quad (\lambda \neq 0)$$

Question 4 solution

Correct answer: C

45°

Use normals (1,0,0) and (1,1,0), with the absolute dot product.

1 Reasoning. Use normals (1,0,0) and (1,1,0), with the absolute dot product.

$$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

Question 5 solution

Correct answer: B

A line

Two independent constraints leave one free parameter.

1 Reasoning. Two independent constraints leave one free parameter.

$$z = t, \quad y = 1 + t, \quad x = 1 - t$$

Question 6 solution

Correct answer: B

5

The line is vertical. The closest point has the same z-coordinate as P, so only the x- and y-displacements contribute.

1 Reasoning. The line is vertical. The closest point has the same z-coordinate as P, so only the x- and y-displacements contribute.

$$d = \sqrt{(4-1)^2 + (6-2)^2} = 5$$

Question 7 solution

Correct answer: B

Contained in the plane

Substitution produces an identity for every parameter value.

1 Reasoning. Substitution produces an identity for every parameter value.

$$(1 + t) + 2 - t = 3$$

Question 8 solution

Correct answer: D

Skew

A non-zero scalar triple product means the connector and directions are not coplanar. Non-coplanar lines cannot intersect.

1 Reasoning. A non-zero scalar triple product means the connector and directions are not coplanar. Non-coplanar lines cannot intersect.

$$\vec{c} \cdot (\vec{d_1} \times \vec{d_2}) \neq 0 \Rightarrow \text{non-coplanar}$$

Question 9 solution

Model response

Substitution gives $(2+t)-(-1+2t)+(1-t)=0$, hence $4-2t=0$ and $t=2$. The intersection is $(4,3,-1)$. It equals the line evaluated at 2 and satisfies $4-3-1=0$.

Marking criteria

1. Substitutes all parametric coordinates into the plane (1 mark).
2. Finds t=2 (1 mark).
3. Obtains (4,3,-1) (1 mark).
4. Verifies both line and plane membership (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Substitute. Keep the negative sign before the y-expression.

$$2+t+1-2t+1-t=0 \Rightarrow t=2$$

2 Evaluate and check. Return to the line, then test the plane.

$$P=(4,3,-1), \qquad 4-3-1=0$$

Question 10 solution

Model response

Adding the equations gives $3x=3$, so $x=1$. The second gives $z=y-1$. Set $y=1+t$ to obtain $\vec{r}=(1,1,0)+t(0,1,1)$. The anchor gives 3 and 0 in the respective equations; the direction has dot product zero with both normals.

Marking criteria

1. Eliminates y and z to obtain x=1 (1 mark).
2. Finds the relation z=y-1 (1 mark).
3. Writes a valid vector equation (1 mark).
4. Checks both the anchor and direction (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Eliminate. Add the equations.

$$3x=3 \Rightarrow x=1, \qquad 1-y+z=0$$

2 Parameterize. Let y=1+t.

$$\vec{r}=(1,1,0)+t(0,1,1)$$

3 Verify direction. Test both normals.

$$(2,1,-1)\cdot(0,1,1)=0, \quad (1,-1,1)\cdot(0,1,1)=0$$

Question 11 solution

Model response

Subtracting the second equation from the first gives $y=2$. Then $x+z=4$ and $2x-z=1$. Adding gives $x=5/3$, and $z=7/3$. The three planes have exactly one common point, $(5/3,2,7/3)$. Its substitutions give 6, 2, and 3.

Marking criteria

1. Eliminates to obtain y=2 (1 mark).
2. Solves the remaining independent equations (1 mark).
3. Finds the unique point (5/3,2,7/3) (1 mark).
4. Verifies it in all three planes and states uniqueness (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Eliminate. Use the first two equations.

$$2y=4 \Rightarrow y=2$$

2 Solve. Substitute y into the remaining equations.

$$x+z=4, \quad 2x-z=1 \Rightarrow x=\frac{5}{3},\ z=\frac{7}{3}$$

3 Verify. Evaluate all left-hand sides.

$$\frac{5}{3}+2+\frac{7}{3}=6, \quad \frac{5}{3}-2+\frac{7}{3}=2, \quad \frac{10}{3}+2-\frac{7}{3}=3$$

Question 12 solution

Model response

The connector is $\overrightarrow{AP}=(3,2,2)$, and $\overrightarrow{AP}\times\vec{d}=(0,-4,4)$. Thus $d=\sqrt{32}/3=4\sqrt{2}/3$. The cross-product magnitude is the area of a parallelogram whose base has length $\|\vec{d}\|$; dividing by that base gives the perpendicular height.

Marking criteria

1. Forms AP=(3,2,2) (1 mark).
2. Computes AP cross d=(0,-4,4) (1 mark).
3. Obtains exact distance 4 square root of 2 divided by 3 (1 mark).
4. Explains the parallelogram area divided by base length (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Connector and cross product. Use the line anchor A.

$$(3,2,2)\times(1,2,2)=(0,-4,4)$$

2 Distance. Divide area by base length.

$$d=\frac{\sqrt{0^2+(-4)^2+4^2}}{\sqrt{1+4+4}}=\frac{4\sqrt{2}}{3}$$

Question 13 solution

Model response

With anchor $A=(-1,2)$, the connector is $\overrightarrow{AP}=(5,-1)$. The foot parameter is $t=\frac{(5,-1)\cdot(2,1)}{(2,1)\cdot(2,1)}=9/5$, giving $H=(13/5,19/5)$. Thus $P-H=(7/5,-14/5)$ and $PH=7\sqrt{5}/5$. Its dot product with $(2,1)$ is zero, so it is the perpendicular connector and gives the shortest distance.

Marking criteria

1. Finds the foot parameter t=9/5 using projection (1 mark).
2. Obtains H=(13/5,19/5) (1 mark).
3. Obtains the exact distance 7 square root of 5 divided by 5 (1 mark).
4. Shows PH is perpendicular to the line and explains that perpendicular separation is shortest (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Project the connector. The projection coefficient gives the line parameter of the perpendicular foot.

$$t=\frac{(5,-1)\cdot(2,1)}{(2,1)\cdot(2,1)}=\frac{9}{5}$$

2 Find the foot and distance. Evaluate the line at the foot parameter, then measure the connector.

$$H=\left(\frac{13}{5},\frac{19}{5}\right),\qquad PH=\sqrt{\frac{49}{25}+\frac{196}{25}}=\frac{7\sqrt{5}}{5}$$

3 Verify perpendicularity. A zero dot product establishes the right angle.

$$\left(\frac{7}{5},-\frac{14}{5}\right)\cdot(2,1)=0$$

Question 14 solution

Model response

With $\vec{n} = (1, 2, 2)$, $\vec{d} \cdot \vec{n} = 8$, and both magnitudes equal 3. Thus $\theta = \sin^{-1}(8/9) \approx 62.7°$. The dot-product cosine initially describes the angle with the normal. The line-plane angle is its complement, so its sine equals that cosine ratio.

Marking criteria

1. Uses the plane normal (1,2,2) and dot product 8 (1 mark).
2. Finds both magnitudes as 3 (1 mark).
3. Obtains arcsin(8/9), approximately 62.7 degrees (1 mark).
4. Explains that the line-plane angle complements the angle with the normal (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Dot product. Use the plane normal.

$$(2, 1, 2) \cdot (1, 2, 2) = 8, \qquad \|\vec{d}\| = \|\vec{n}\| = 3$$

2 Line-plane angle. Use the complementary angle relation.

$$\sin\theta = \frac{8}{9} \Rightarrow \theta = \sin^{-1}\left(\frac{8}{9}\right) \approx 62.7°$$

Question 15 solution

Model response

If $k = 2$, the second left-hand side is twice the first, but $3 \neq 2$, so the planes are parallel and distinct. If $k \neq 2$, subtracting twice the first equation gives $(k-2)z = 1$. Therefore $z = 1/(k-2)$ and $x + y = (k-3)/(k-2)$. One line is $\vec{r} = ((k-3)/(k-2), 0, 1/(k-2)) + t(1, -1, 0)$.

Marking criteria

1. Identifies k=2 as the parallel case (1 mark).
2. Shows that k=2 gives distinct planes with no intersection (1 mark).
3. Finds z=1/(k-2) when k is not 2 (1 mark).
4. Gives a correct intersection line for k not equal to 2 (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Eliminate. Subtract twice the first equation.

$$(k-2)z = 1$$

2 Parallel case. At k=2 the system is contradictory.

$$k = 2 \Rightarrow 0 = 1$$

3 Intersecting case. Let y=-t when k is not 2.

$$\vec{r} = \left(\frac{k-3}{k-2}, 0, \frac{1}{k-2}\right) + t(1, -1, 0)$$

Question 16 solution

Model response

Substitution gives $(1 + t) - t + 1 = 2$, an identity for every real t. It does not force $t = 0$. The line is contained in the plane, so there are infinitely many common points and $L \cap \Pi = \{(1 + t, -t, 1) : t \in \mathbb{R}\}$.

Marking criteria

1. Shows the substitution is an identity (1 mark).
2. Explains that no restriction on t remains (1 mark).
3. States that every point of the line lies in the plane (1 mark).
4. Gives the full intersection set with all real t (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Substitute. Simplify without assigning a parameter value.

$$(1 + t) - t + 1 = 2 \Rightarrow 2 = 2$$

2 Interpret. An identity is true for every real parameter.

$$L \subset \Pi, \qquad L \cap \Pi = L$$

6 Final examination

6.1 Lines and Planes final examination

A 60-mark examination on Sections 8.1 to 8.6 and 9.1 to 9.6, combining representation, sketching, intersections, systems, exact distances, and mathematical communication.

24 questions 60 marks Suggested time: 100 min

1. Answer every question. Each question shows its mark allocation.
2. For written responses, show algebraic steps and justify geometric conclusions using vectors or equations.
3. Give exact values unless an approximation is requested. Give approximate angles to the nearest tenth of a degree.
4. Use distinct parameters for different lines. Include a check whenever the question requests verification.
5. Written responses are self-assessed against the marking criteria. They are not included in the automatic multiple-choice score.

Question paper

Section A: Multiple choice

Question 1 [1 mark]

Categories: K
Which statement about a Cartesian line in R³ is correct?

A. One equation Ax+By+Cz+D=0 always defines a line.

B. It can be represented as the intersection of two non-parallel planes.

C. It requires three independent parameters.

D. It has no direction vector.

Question 2 [1 mark]

Categories: K, C
Convert $\frac{x-2}{3} = \frac{y+1}{-2} = z - 4$ to parametric form.

A. $x = 2 + 3t,\ y = -1 - 2t,\ z = 4 + t$

B. $x = 3 + 2t,\ y = -2 - t,\ z = 1 + 4t$

C. $x = 2 - 3t,\ y = -1 + 2t,\ z = 4 + t$

D. $x = 2 + 3t,\ y = 1 - 2t,\ z = 4$

Question 3 [1 mark]

Categories: A
Which plane contains A(1,0,0), B(0,1,0), and C(0,0,1)?

A. $x + y + z = 0$

B. $x + y + z = 1$

C. $x - y + z = 1$

D. $xyz = 1$

Question 4 [1 mark]

Categories: A
Find the intersection of $\vec{r} = (0, 1, 0) + t(2, -1, 3)$ and $x + y - z = 0$.

A. $(0, 1, 0)$

B. $(2, 0, 3)$

C. $(1, 1/2, 3/2)$

D. No intersection

Question 5 [1 mark]

Categories: T
Classify $2x - y + z = 4$ and $4x - 2y + 2z = 11$.

A. Intersect in a line

B. Coincident

C. Distinct parallel planes

D. Perpendicular planes

Question 6 [1 mark]

Categories: T, C
Which line is the intersection of $x + y + z = 3$ and $x - y + z = 1$?

A. $\vec{r} = (2, 1, 0) + t(-1, 0, 1)$

B. $\vec{r} = (0, 0, 0) + t(1, 1, 1)$

C. $\vec{r} = (3, 1, 1) + t(1, 0, 1)$

D. $\vec{r} = (2, 0, 1) + t(0, 1, 0)$

Question 7 [1 mark]

Categories: A, T
Find the distance from $P(3, 2, 0)$ to $\vec{r} = (1, 1, 1) + t(2, 0, -1)$.

A. 0

B. 1

C. $\sqrt{5}$

D. 5

Question 8 [1 mark]

Categories: C
Substituting a line into a plane simplifies to 0=0. Which conclusion communicates the result completely?

A. The equations are wrong.

B. The line is parallel to the plane, so there is no intersection.

C. Every point on the line satisfies the plane; the line lies in the plane and there are infinitely many intersections.

D. The plane lies on the line.

Question 9 [1 mark]

Categories: K, A
Which three points should be plotted first to sketch $3x + 2y + 6z = 6$ by intercepts?

A. $(2, 0, 0), (0, 3, 0), (0, 0, 1)$

B. $(3, 0, 0), (0, 2, 0), (0, 0, 6)$

C. $(6, 0, 0), (0, 6, 0), (0, 0, 6)$

D. $(1, 0, 0), (0, 1, 0), (0, 0, 1)$

Question 10 [1 mark]

Categories: A
Find the distance from $P(4, 1)$ to $3x - 4y + 2 = 0$.

A. 2

B. $\frac{10}{7}$

C. $\sqrt{5}$

D. 10

Question 11 [1 mark]

Categories: K, A
In $\mathbb{R}^2$, which parametric equations represent $\vec{r} = (2, -1) + t(3, 2)$?

A. $x = 2 + 3t,\ y = -1 + 2t$

B. $x = 3 + 2t,\ y = 2 - t$

C. $x = 2 + 3t,\ y = 1 + 2t$

D. $x = 2 + 3s,\ y = -1 + 2t$, with independent s, t

Question 12 [1 mark]

Categories: K, T
Which statement gives the correct sketching information for the plane $x + y = 0$ in $\mathbb{R}^3$?

A. It is the coordinate plane z=0.

B. It is only a line in the xy-plane.

C. It contains the z-axis, and three distinct non-zero axis intercepts cannot be used to sketch it.

D. It has the three intercepts (1,0,0), (0,1,0), and (0,0,1).

Question 13 [1 mark]

Categories: K, T
Elimination in a system of three linear equations produces the equation $0 = 4$. What does this establish?

A. The system has one solution.

B. The solution set is a line.

C. The solution set is a plane.

D. The system is inconsistent and has no common solution.

Question 14 [1 mark]

Categories: T
The planes are $x + y + z = 2$, $2x - y + z = 1$, and $3x + 2z = 3$. What is their common solution set?

A. Exactly one point

B. A line

C. A plane

D. No common points

Section B: Application

Question 15 [5 marks]

Categories: A, T, C
Two lines intersect at $P = (1, 0, 2)$ and have directions $\vec{d_1} = (1, 1, 0)$ and $\vec{d_2} = (0, 1, 1)$. Construct a Cartesian equation of their plane and verify the result.
Show your working and state a complete conclusion.

Question 16 [4 marks]

Categories: A, C
For the line $3x - 4y + 5 = 0$ in $\mathbb{R}^2$, (a) find an anchor and a direction vector, (b) write vector and parametric equations, and (c) find its exact distance from the origin.
Show your working and state a complete conclusion.

Question 17 [4 marks]

Categories: A, C
A line passes through $A(2, -1, 3)$ and $B(2, 3, 1)$. Write vector and symmetric equations, handling the zero direction component correctly. Then determine whether $P(2, 5, 0)$ lies on the line.
Show your working and state a complete conclusion.

Question 18 [4 marks]

Categories: A, T, C
The plane is $\vec{r} = (0, 1, -1) + s(1, 0, 2) + t(0, 2, -1)$. (a) Write its parametric equations. (b) Find a Cartesian equation. (c) Explain why the two spans define a plane rather than a line.
Show your working and state a complete conclusion.

Question 19 [4 marks]

Categories: A, C
Describe how to sketch $2x + 3z = 6$ in $\mathbb{R}^3$. Give the x- and z-intercepts, explain why there is no y-intercept, and identify the direction in which the plane extends. Give one additional point that helps show this extension.
Show your working and state a complete conclusion.

Question 20 [5 marks]

Categories: A, T, C
Determine the relationship between $L_1 : \vec{r} = (1,-1,2) + t(2,1,-1)$ and $L_2 : \vec{r} = (2,4,-3) + s(1,-1,1)$. Find any intersection, verify all coordinates, and state a complete geometric conclusion.
Show your working and state a complete conclusion.

Question 21 [5 marks]

Categories: A, T, C
Solve the system $x + y + z = 6$, $2x - y + z = 3$, and $x + 2y - z = 2$ by elimination. Show your reduced equations, verify the solution in all three original equations, and interpret the result as an intersection of planes.
Show your working and state a complete conclusion.

Question 22 [5 marks]

Categories: A, T, C
Find the exact distance from $P(4,1,2)$ to the plane $2x - y + 2z = 3$. Also find the perpendicular foot H, then verify that H lies on the plane and that $\overrightarrow{HP}$ is normal to it.
Show your working and state a complete conclusion.

Section C: Thinking and Communication

Question 23 [5 marks]

Categories: K, T, C
A line $\vec{r} = (0,0,0) + t(1,-1,0)$ is compared with the plane $x + y + z = 2$. Explain the relationship using both a vector test and a geometric conclusion.
Show your working and state a complete conclusion.

Question 24 [5 marks]

Categories: T, C
For all real values of k, classify the system $x + y + z = 3$, $x - y + z = 1$, and $2x + ky + 2z = 5$. Give the complete common solution set when one exists, and explain whether a unique solution is possible.
Show your working and state a complete conclusion.

Worked solutions and marking guidance

Use the question number to match each solution to the paper above.

Question 1 solution

Correct answer: B

It can be represented as the intersection of two non-parallel planes.

One scalar linear equation in R³ defines a plane; two independent plane equations can intersect in a line.

1 Dimension check. Each independent scalar constraint reduces dimension by one.

$$\mathbb{R}^3 \xrightarrow{\text{one constraint}} \text{plane} \xrightarrow{\text{second constraint}} \text{line}$$

Question 2 solution

Correct answer: A

$x = 2 + 3t,\ y = -1 - 2t,\ z = 4 + t$

Set every symmetric expression equal to the shared parameter t. Since z-4=t, z=4+t.

1 Set equal to t. Solve each relation.

$$\frac{x-2}{3} = t,\ \frac{y+1}{-2} = t,\ z - 4 = t$$

Question 3 solution

Correct answer: B

$x + y + z = 1$

Each point is a unit intercept, so the intercept equation is x+y+z=1.

1 Spans. Use AB=(-1,1,0) and AC=(-1,0,1).

$$\overrightarrow{AB} \times \overrightarrow{AC} = (1, 1, 1)$$

2 Plane. Use A in point-normal form.

$$(x - 1) + y + z = 0 \Rightarrow x + y + z = 1$$

Question 4 solution

Correct answer: C

$(1, 1/2, 3/2)$

Substitution gives 2t+(1-t)-3t=0, so 1-2t=0 and t=1/2. Evaluating the line gives (1,1/2,3/2).

1 Substitute. Use x=2t, y=1-t, z=3t.

$$2t + 1 - t - 3t = 0 \Rightarrow 1 - 2t = 0 \Rightarrow t = \frac{1}{2}$$

2 Evaluate. Return to the line.

$$P = \left(1, \frac{1}{2}, \frac{3}{2}\right)$$

Question 5 solution

Correct answer: C

Distinct parallel planes

The normals scale by 2, but 2(4)=8≠11. The planes are parallel and distinct.

1 Compare scale. Double the first equation's left side and constant.

$$2(2x - y + z) = 4x - 2y + 2z, \qquad 2(4) = 8 \neq 11$$

Question 6 solution

Correct answer: A

$\vec{r} = (2, 1, 0) + t(-1, 0, 1)$

Subtracting gives y=1; then x+z=2. Let z=t to obtain the listed line.

1 Eliminate. Subtract the equations.

$$2y = 2 \Rightarrow y = 1$$

2 Parameterize. Let z=t in x+z=2.

$$x = 2 - t \Rightarrow \vec{r} = (2, 1, 0) + t(-1, 0, 1)$$

Question 7 solution

Correct answer: B

1

AP=(2,1,-1), AP×d=(-1,0,-2), and both the numerator and denominator have magnitude √5.

1 Apply formula. Use the cross-product distance formula.

$$d = \frac{\|(2, 1, -1) \times (2, 0, -1)\|}{\|(2, 0, -1)\|} = \frac{\|(-1, 0, -2)\|}{\sqrt{5}} = 1$$

Question 8 solution

Correct answer: C

Every point on the line satisfies the plane; the line lies in the plane and there are infinitely many intersections.

An identity means the equation is true for every parameter value, not just one.

1 Interpret the identity. No restriction remains on t.

$$0 = 0 \text{ for all } t \in \mathbb{R} \Rightarrow L \subset \Pi$$

Question 9 solution

Correct answer: A

$(2,0,0), (0,3,0), (0,0,1)$

Set the other two variables to zero for each axis intercept.

1 Find the intercepts. Solve each one-variable equation.

$$x = 2, \qquad y = 3, \qquad z = 1$$

Question 10 solution

Correct answer: A

2

Substitute the point into the numerator and divide by the magnitude of the normal.

1 Use the R² formula. Evaluate the numerator and denominator.

$$d = \frac{|3(4) - 4(1) + 2|}{\sqrt{3^2 + (-4)^2}} = \frac{10}{5} = 2$$

Question 11 solution

Correct answer: A

$x = 2 + 3t,\ y = -1 + 2t$

Each coordinate uses the same parameter. The starting point supplies the constants and the direction supplies the parameter coefficients.

1 Reasoning. Each coordinate uses the same parameter. The starting point supplies the constants and the direction supplies the parameter coefficients.

$$\vec{r} = (2 + 3t, -1 + 2t)$$

Question 12 solution

Correct answer: C

It contains the z-axis, and three distinct non-zero axis intercepts cannot be used to sketch it.

The variable z is unrestricted. Every point (0,0,z) satisfies the equation, so the whole z-axis lies in the plane. Both x- and y-axis intersections occur at the origin.

1 Reasoning. The variable z is unrestricted. Every point (0,0,z) satisfies the equation, so the whole z-axis lies in the plane. Both x- and y-axis intersections occur at the origin.

$$x = y = 0 \Rightarrow 0 = 0 \quad \text{for every } z \in \mathbb{R}$$

Question 13 solution

Correct answer: D

The system is inconsistent and has no common solution.

A contradiction cannot hold for any values of the variables. Since elimination preserves the solution set, the original equations have no simultaneous solution.

1 Reasoning. A contradiction cannot hold for any values of the variables. Since elimination preserves the solution set, the original equations have no simultaneous solution.

$$0 = 4 \quad \text{is false} \Rightarrow \text{no solution}$$

Question 14 solution

Correct answer: B

A line

The third equation is the sum of the first two. Their normals are not parallel, so the two independent constraints determine a line; the third adds no restriction.

1 Reasoning. The third equation is the sum of the first two. Their normals are not parallel, so the two independent constraints determine a line; the third adds no restriction.

$$(x + y + z) + (2x - y + z) = 3x + 2z = 3$$

Question 15 solution

Model response

The lines provide two in-plane directions. Their cross product is (1,1,0) cross (0,1,1)=(1,-1,1), so a normal is n=(1,-1,1). Using P gives (x-1)-y+(z-2)=0, or x-y+z=3. The point P satisfies the equation, and the normal has dot product 0 with each line direction, so the plane is verified.

Marking criteria

1. Uses the intersection point as the plane anchor (1 mark).
2. Finds a non-zero normal using the cross product (1 mark).
3. Builds and simplifies the Cartesian equation (1 mark).
4. Verifies that the anchor satisfies the equation (1 mark).
5. Verifies that both line directions are perpendicular to the normal (1 mark).

Two intersecting lines determine a plane when their direction vectors are not parallel.

1 Find the normal. Cross the line directions.

$$(1, 1, 0) \times (0, 1, 1) = (1, -1, 1)$$

2 Use the intersection point. Build point-normal form.

$$(x - 1) - y + (z - 2) = 0 \Rightarrow x - y + z = 3$$

3 Verify. Check the anchor and perpendicularity.

$$(1,-1,1)\cdot(1,1,0)=0, \quad (1,-1,1)\cdot(0,1,1)=0$$

Question 16 solution

Model response

A normal is $(3,-4)$, so choose $\vec{d}=(4,3)$. The point $(1,2)$ lies on the line because $3-8+5=0$. Thus $\vec{r}=(1,2)+t(4,3)$, with $x=1+4t$, $y=2+3t$, $t\in\mathbb{R}$. The distance from the origin is $d=|5|/\sqrt{3^2+(-4)^2}=1$.

Marking criteria

1. Chooses a direction perpendicular to (3,-4), such as (4,3) (1 mark).
2. Finds an anchor satisfying the equation, such as (1,2) (1 mark).
3. Writes correct vector and parametric equations with one shared parameter (1 mark).
4. Calculates the exact origin-to-line distance as 1 (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Anchor and direction. An anchor satisfies the equation; a direction is perpendicular to its normal.

$$3(1)-4(2)+5=0, \qquad (3,-4)\cdot(4,3)=0$$

2 Represent the line. Use one real parameter for both coordinates.

$$\vec{r}=(1,2)+t(4,3), \qquad x=1+4t, \quad y=2+3t$$

3 Distance. Substitute the origin into the scalar distance formula.

$$d=\frac{|3(0)-4(0)+5|}{\sqrt{3^2+(-4)^2}}=1$$

Question 17 solution

Model response

The displacement is $B-A=(0,4,-2)$, so choose $\vec{d}=(0,2,-1)$. The vector equation is $\vec{r}=(2,-1,3)+t(0,2,-1)$. Symmetric form is $x=2$, $\frac{y+1}{2}=\frac{z-3}{-1}$. For $t=3$, the line gives $(2,5,0)$, so P lies on it.

Marking criteria

1. Finds a valid direction such as (0,2,-1) (1 mark).
2. Writes a correct vector equation (1 mark).
3. Writes x=2 separately and equates only the non-zero direction ratios (1 mark).
4. Uses the same parameter in all coordinates to verify that P lies on the line (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Direction. Simplify the displacement without changing the line.

$$B - A = (0, 4, -2) = 2(0, 2, -1)$$

2 Equation forms. A zero direction component gives a fixed coordinate, not division by zero.

$$\vec{r} = (2, -1, 3) + t(0, 2, -1), \qquad x = 2, \quad \frac{y+1}{2} = \frac{z-3}{-1}$$

3 Membership. Substitute one shared parameter.

$$t = 3 \Rightarrow (2, -1 + 6, 3 - 3) = (2, 5, 0)$$

Question 18 solution

Model response

The parametric equations are $x = s$, $y = 1 + 2t$, and $z = -1 + 2s - t$. The cross product of the spans is $(-4, 1, 2)$, so a Cartesian equation is $-4x + y + 2z = -1$. The cross product is non-zero, which proves that the spans are not parallel. They provide two independent directions and therefore define a plane.

Marking criteria

1. Writes all three parametric equations correctly (1 mark).
2. Finds a non-zero normal (-4,1,2), or a scalar multiple (1 mark).
3. Obtains -4x+y+2z=-1, or an equivalent equation (1 mark).
4. Uses the non-zero cross product to explain span independence and the plane dimension (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Parametric form. Read each coordinate separately.

$$x = s, \qquad y = 1 + 2t, \qquad z = -1 + 2s - t$$

2 Normal. Cross the two spanning vectors.

$$(1, 0, 2) \times (0, 2, -1) = (-4, 1, 2) \neq \vec{0}$$

3 Cartesian form. Use the anchor (0,1,-1) in point-normal form.

$$-4x + (y - 1) + 2(z + 1) = 0 \Rightarrow -4x + y + 2z = -1$$

Question 19 solution

Model response

The x-intercept is $(3,0,0)$ and the z-intercept is $(0,0,2)$. A y-axis point has $x = z = 0$, which would require $0 = 6$, so there is no y-intercept. Since y does not appear, draw the segment between the intercepts and extend it in both y-directions to represent the infinite plane. For example, $(3,1,0)$ also lies on the plane.

Marking criteria

1. Finds the intercepts (3,0,0) and (0,0,2) (1 mark).
2. Explains that setting x=z=0 gives a contradiction, so no y-axis point lies on the plane (1 mark).
3. States that y is unrestricted and the plane extends in the y-direction (1 mark).
4. Gives a valid additional point such as (3,1,0) and describes extending the intercept segment parallel to the y-axis (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Intercepts. Set the other coordinates to zero.

$$z = 0 \Rightarrow x = 3, \qquad x = 0 \Rightarrow z = 2$$

2 Missing intercept. Test the entire y-axis.

$$x = z = 0 \Rightarrow 0 = 6 \quad \text{which is impossible}$$

3 Extend the sketch. The y-coordinate is free.

$$(3, y, 0) \in \Pi \quad \text{for all } y \in \mathbb{R}$$

Question 20 solution

Model response

The x- and y-equations give $2t - s = 1$ and $t + s = 5$. Their solution is $t = 2$, $s = 3$, giving $(5,1,0)$. The z-coordinates agree because $2 - 2 = -3 + 3 = 0$. The directions are not scalar multiples, so the lines intersect at exactly one point, $(5,1,0)$.

Marking criteria

1. Equates coordinates using distinct parameters (1 mark).
2. Finds t=2 and s=3 (1 mark).
3. Finds the point (5,1,0) (1 mark).
4. Checks the third coordinate as well as the coordinates used to solve (1 mark).
5. Uses the non-parallel directions to conclude that the lines intersect at exactly one point (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Equate coordinates. Keep the two parameters independent.

$$1+2t=2+s, \qquad -1+t=4-s, \qquad 2-t=-3+s$$

2 Solve. Add the first two simplified equations.

$$2t-s=1, \quad t+s=5 \Rightarrow 3t=6 \Rightarrow t=2,\ s=3$$

3 Verify and classify. Both lines produce the same three coordinates, and their directions are not parallel.

$$\vec{r}_1(2)=\vec{r}_2(3)=(5,1,0)$$

Question 21 solution

Model response

Subtracting twice the first equation from the second gives $3y+z=9$. Subtracting the first from the third gives $y-2z=-4$. Solving gives $y=2$, $z=3$, then $x=1$. The left sides of the original equations become 6, 3, and 2. Thus the system is consistent with the unique solution $(1,2,3)$, and the three planes meet at this point.

Marking criteria

1. Eliminates x to obtain 3y+z=9 (1 mark).
2. Obtains the independent equation y-2z=-4 (1 mark).
3. Solves y=2 and z=3 (1 mark).
4. Finds x=1 (1 mark).
5. Checks all three original equations and identifies one common point (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Eliminate x. Use equation 1 to eliminate x from equations 2 and 3.

$$E_2-2E_1:\ -3y-z=-9, \qquad E_3-E_1:\ y-2z=-4$$

2 Solve the reduced system. Substitute z=9-3y into the second reduced equation.

$$y-2(9-3y)=-4 \Rightarrow 7y=14 \Rightarrow y=2,\ z=3,\ x=1$$

3 Verify. Check all three original equations.

$$1+2+3=6, \qquad 2(1)-2+3=3, \qquad 1+2(2)-3=2$$

Question 22 solution

Model response

The normal is $\vec{n} = (2, -1, 2)$, with magnitude 3. The residual at P is $2(4) - 1 + 2(2) - 3 = 8$, so $d = 8/3$. The foot is $H = P - \frac{8}{9}\vec{n} = (20/9, 17/9, 2/9)$. It satisfies $2(20/9) - 17/9 + 2(2/9) = 3$. Also $\overrightarrow{HP} = P - H = \frac{8}{9}(2, -1, 2)$, so the connector is perpendicular to the plane.

Marking criteria

1. Calculates the signed residual as 8 (1 mark).
2. Uses the normal (2,-1,2) with magnitude 3 (1 mark).
3. Obtains the exact distance 8/3 (1 mark).
4. Finds H=(20/9,17/9,2/9) using the normal projection (1 mark).
5. Verifies plane membership and that HP is a scalar multiple of the normal (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 Distance. Divide the absolute residual by the normal magnitude.

$$d = \frac{|2(4) - 1 + 2(2) - 3|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{8}{3}$$

2 Perpendicular foot. Remove the normal component from P.

$$H = (4, 1, 2) - \frac{8}{9}(2, -1, 2) = \left(\frac{20}{9}, \frac{17}{9}, \frac{2}{9}\right)$$

3 Verify. Check both membership and the normal direction of the connector.

$$2\left(\frac{20}{9}\right) - \frac{17}{9} + 2\left(\frac{2}{9}\right) = 3, \qquad P - H = \frac{8}{9}\vec{n}$$

Question 23 solution

Model response

The plane normal is n=(1,1,1). The dot product n dot d is 1-1+0=0, so the line direction is parallel to the plane. The anchor (0,0,0) does not satisfy x+y+z=2. Therefore the entire line is outside the plane, the line is parallel to it, and there are no intersection points.

Marking criteria

1. Identifies the plane normal (1 mark).
2. Calculates the normal-direction dot product as zero (1 mark).
3. Explains that this establishes a direction parallel to the plane (1 mark).
4. Checks that the line anchor is outside the plane (1 mark).
5. Concludes that the line is parallel and disjoint, with no intersection (1 mark).

A zero normal-direction dot product gives parallel direction. The anchor test then separates a contained line from a disjoint one.

1 Vector test. Use the plane normal and line direction.

$$(1,1,1)\cdot(1,-1,0)=0$$

2 Anchor test. Evaluate the plane at the origin.

$$0+0+0\neq 2$$

3 Conclusion. The line is parallel and disjoint.

$$L\parallel\Pi,\qquad L\cap\Pi=\varnothing$$

Question 24 solution

Model response

Subtracting the second equation from the first gives $y=1$, and then $x+z=2$. The third equation becomes $2(x+z)+k=5$, or $4+k=5$. For $k=1$, it adds no new restriction, so the solution set is $\vec{r}=(2,1,0)+t(-1,0,1)$, $t\in\mathbb{R}$. For $k\neq 1$, the third equation contradicts the first two, so the system is inconsistent. No value of k produces a unique point.

Marking criteria

1. Uses the first two equations to find y=1 (1 mark).
2. Obtains x+z=2 (1 mark).
3. Derives the compatibility condition k=1 from the third equation (1 mark).
4. Gives the intersection line r=(2,1,0)+t(-1,0,1) when k=1 (1 mark).
5. Concludes that k not equal to 1 is inconsistent and that no k gives a unique point (1 mark).

A complete response shows the method, gives an exact result, and explains its geometric meaning.

1 First two planes. Subtract, then substitute y=1.

$$2y=2\Rightarrow y=1,\qquad x+z=2$$

2 Third-plane condition. Its restriction is independent of the free line parameter.

$$2(x+z)+ky=5\Rightarrow 4+k=5\Rightarrow k=1$$

3 Classify. Let z=t in the consistent case.

$$k=1:\ \vec{r}=(2,1,0)+t(-1,0,1),\ t\in\mathbb{R};\qquad k\neq 1:\ \varnothing$$

7 Formula reference

Lines

Vector

$\vec{r} = \vec{r}_0 + t\vec{d}$

d ≠ 0

Parametric

$x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$

one shared parameter

Symmetric

$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

separate zero-direction coordinates

R² scalar

$Ax + By + C = 0$

normal (A,B)

Planes

Vector

$\vec{r} = \vec{r}_0 + s\vec{a} + t\vec{b}$

a and b independent

Point-normal

$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$

n perpendicular to plane

Cartesian

$Ax + By + Cz + D = 0$

normal (A,B,C)

Normal from spans

$\vec{n} = \vec{a} \times \vec{b}$

cross product non-zero

Intersections and distance

Line-plane parameter

$t = \frac{k - \vec{n} \cdot \vec{r}_0}{\vec{n} \cdot \vec{d}}$

when n·d ≠ 0

Two-plane direction

$\vec{d} = \vec{n}_1 \times \vec{n}_2$

non-parallel normals

Point-line R²

$d = \frac{|Ax_P + By_P + C|}{\sqrt{A^2 + B^2}}$

scalar line form

Point-line R³

$d = \frac{\|\overrightarrow{AP} \times \vec{d}\|}{\|\vec{d}\|}$

A lies on line

Angles and system checks

Line-plane angle

$\sin\theta = \frac{|\vec{d} \cdot \vec{n}|}{\|\vec{d}\|\|\vec{n}\|}$

acute angle from line to plane

Plane-plane angle

$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\|\vec{n}_2\|}$

acute angle between normals

Point-plane distance

$d = \frac{|Ax_P + By_P + Cz_P + D|}{\sqrt{A^2 + B^2 + C^2}}$

plane in Cartesian form

Consistency

$0 = 0 \Rightarrow$ redundant equation, $\quad 0 = c \neq 0 \Rightarrow$ no solution

a consistent system with a free variable has infinitely many solutions

8 Classroom sources and online resources

The lesson sequence follows Sections 8.1 to 8.6 and 9.1 to 9.6 identified in the supplied classroom materials. Explanations, worked examples, diagrams and assessment solutions in this coursebook are typed course content. The classroom slide page references in the lessons use PDF page numbering.

1. Chapter8 Slides(1).pdf. Classroom slide deck, 58 PDF pages. Section 8.6 is listed in the chapter objectives; the coursebook includes the required plane-sketching lesson.
2. 09 - Chapter 9, Relationships Between Points Lines and Planes(1).pdf. Classroom slide deck, 54 PDF pages.
3. CCT Guidlines July 2026 (1)(1).pdf. MCV4U Course Culminating Task Guidelines. Topic H identifies Lines and Planes. The required submission includes the website link and a PDF version.
4. CCT Rubric 2026.pdf. Assessment criteria for accuracy, sequencing, homework, explanation, organization, delivery, assessments and project conference.

Original teaching video recordings are prepared and uploaded by the group. At least two instructional videos must each run for eight minutes or longer. The website provides access to the recordings as they are published.

Online website

https://mcv4u-lines-and-planes.hozantaxprivate.chatgpt.site